

Differential Calculus

Vesa Kaarnioja
vesa.kaarnioja@lut.fi

LUT School of Engineering Sciences

Week 1
October 28–29, 2025

Practical Matters

- **Responsible teacher:** Vesa Kaarnioja (vesa.kaarnioja@lut.fi).
- All course material will be available on the course's Moodle page.
- To complete the course, you must:
 1. Pass the course exam **and**
 2. Complete the mathematics exercise quota (earning a cumulative 30% of mathematics exercise points)[†], **and**
 3. Complete the MATLAB exercise quota (20 MATLAB assignments successfully solved).
- The course consists of weekly lectures (3h+3h), mathematics exercise groups (2h), and MATLAB exercise groups (2h).
- Please register for the exercise groups on a weekly basis on the course Moodle page (under the “Week 1”, “Week 2”, “Week 3”, . . . , tabs).
- Mathematics exercises start already the first week, MATLAB exercises start on the second week.

[†]If you have already completed the MATLAB quota in 2024 or earlier, you can directly attend the exam without retaking the mathematics/MATLAB exercises.

Lectures

- This course's lectures will be held
Tuesdays and **Wednesdays** at **8.30–11.00** in **room 1318**.
- The final lecture on **December 10** will be held in **Viipuri hall**.

Exercise Practices

- This course uses an **exercise quota**: in order to pass the course, students must complete a certain number of exercises (including *both* mathematics exercises *and* MATLAB exercises). You may obtain extra points for the exam by completing additional mathematics exercises beyond the required quota.
- The weekly exercise problems must be solved before the exercise sessions. At the beginning of each session, a name list is passed around, and each student marks the problems they are prepared to present in front of the class. The instructor will then choose students from the list to present their solutions. Points are awarded only for the problems marked on the list.

Exercise Points

Completing more than 30% of the course's exercises accrues bonus points, which can be used to improve your exam grade according to the following table:

At least 30% of exercise points	completion requirement
At least 40% of exercise points	1 course exam point
At least 50% of exercise points	2 course exam points
At least 60% of exercise points	3 course exam points
At least 70% of exercise points	4 course exam points
At least 80% of exercise points	5 course exam points
At least 90% of exercise points	6 course exam points

(6 course exam points corresponds to one perfectly solved exam question.)

The exercise points are valid for the exams on 15 December 2025, 9 February 2026, and 23 March 2026.

Support for Exercises

- A “tutoring lab” (“laskaripaja”) will be organized by the Department of Computational Engineering, who are happy to assist with this course’s assignments:
 - **Time:** Tuesdays at 12-14 and Thursdays at 14-16 in room Yo-talo 216 (near the guild room).
 - **First meeting:** October 30.

Schedule

Week 1: Register for one of this week's mathematics exercise groups by **Wednesday 8pm (tomorrow)**. Please note that you are expected to solve the problems before the exercise session.

Weeks 2–7: Register to one MATLAB exercise group and one mathematics exercise group each week. The registration for each week's exercise groups closes on **Monday 8pm**. Please note that you are expected to solve the problems before the exercise sessions.

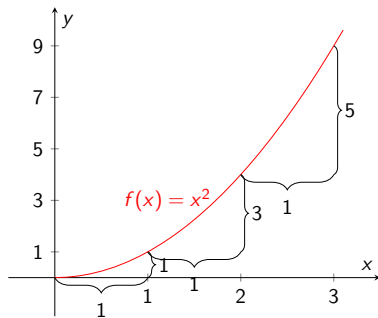
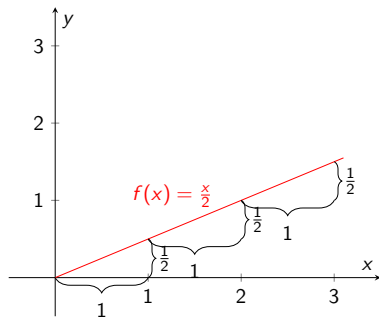
Limit of a Function

Introduction

The limit is a concept that allows for the precise handling of arbitrarily large and small values. This enables the calculation of, for example, the instantaneous rate of change of a function and the area bounded by the graph of the function. However, many concepts based on limits can be formulated in such a way that the limit itself is no longer needed. For this reason, the significance of the limit can easily be misunderstood.

Therefore, the treatment of limits in this document begins with the problem that originally created the need for the concept of a limit and shows how it solved that problem. This concerns the determination of the instantaneous rate of change of a function, which is essentially the derivation of the definition of the derivative.

The rate of change of the function $f(x) = \frac{x}{2}$ is constant. It increases by half a unit as the variable increases by one unit, which is evident in the fact that its graph is a straight line. In contrast, the rate of change of the function $f(x) = x^2$ is not constant; for positive values of the variable, the function increases faster as the value of the variable itself increases. This is illustrated by the steepening of its graph as one moves to the right.



The rate of change of the function $f(x) = x^2$ is thus different for each value of the variable x . Such a value is called the **instantaneous** rate of change of the function. We will henceforth denote the instantaneous rate of change of the function at the point x by $f'(x)$.

Let us attempt to determine the instantaneous rate of change of the function $f(x) = x^2$ at the point $x = 1$. This may seem like a modest goal, but if successful, perhaps the same technique can be applied to an arbitrary point, allowing us to find the instantaneous rate of change at every point at once. This technique could further generalize to other functions as well.

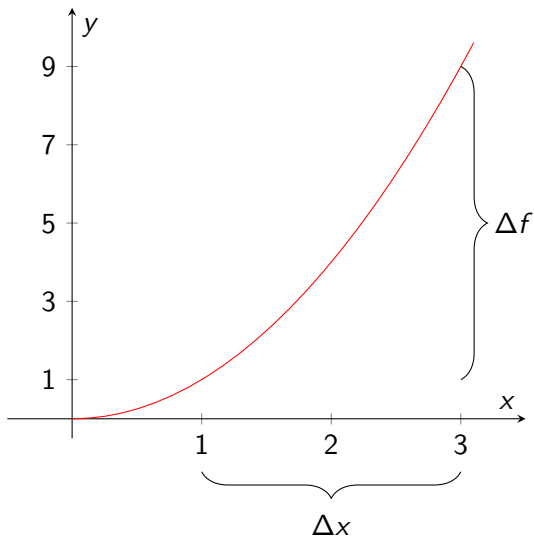
We begin by recalling from physics that the average speed v_k of an object over the interval Δt is calculated using the formula $v_k = \Delta s / \Delta t$, where Δs is the difference between the object's final and initial positions. For example, if an object travels 100 km in two hours, the average speed is given by

$$v_k = \frac{\Delta s}{\Delta t} = \frac{s_2 - s_1}{t_2 - t_1} = \frac{100 - 0}{2 - 0} = \frac{100}{2} = 50 \left(\frac{\text{km}}{\text{h}} \right).$$

Similarly, if the change in the function f over the interval Δx is denoted by Δf , then the average rate of change of the function over that interval is given by the expression $\Delta f / \Delta x$. For example, the average rate of change of the function $f(x) = x^2$ over the interval $[1, 3]$ is

$$\frac{\Delta f}{\Delta x} = \frac{f(3) - f(1)}{3 - 1} = \frac{3^2 - 1^2}{3 - 1} = 4,$$

as illustrated in the accompanying image.



This value can be used as a rough estimate for the instantaneous rate of change of the function f at the point $x = 1$. Therefore, it holds roughly that $f'(x) \approx 4$.

A better estimate is obtained when the upper limit of the interval is brought closer to one. Over the interval $[1, 1.1]$, the corresponding estimate is

$$f'(1) \approx \frac{\Delta f}{\Delta x} = \frac{f(1.1) - f(1)}{1.1 - 1} = \frac{1.1^2 - 1^2}{1.1 - 1} = 2.1.$$

Similarly, over the interval $[1, 1.01]$, the corresponding estimate is

$$f'(1) \approx \frac{\Delta f}{\Delta x} = \frac{f(1.01) - f(1)}{1.01 - 1} = \frac{1.01^2 - 1^2}{1.01 - 1} = 2.01,$$

and over the interval $[1, 1.001]$,

$$f'(1) \approx \frac{\Delta f}{\Delta x} = \frac{f(1.001) - f(1)}{1.001 - 1} = \frac{1.001^2 - 1^2}{1.001 - 1} = 2.001.$$

The estimate appears to converge towards the number 2. However, if we attempt to calculate the exact value by examining the interval $[1, 1]$, we encounter the problem:

$$\frac{\Delta f}{\Delta x} = \frac{f(1) - f(1)}{1 - 1} = \frac{0}{0},$$

which means we would be dividing by zero. This problem can also be understood physically: if no time elapses, then no position can change, so we cannot speak of a rate of change.

If the previous analysis is conducted in symbolic form, we can progress a bit further. In this case, we examine the interval $[1, 1 + h]$, where h is an arbitrary positive number. It could also be negative, but we will stick to the positive for simplicity. For example, in the case where $h = 0.001$, we obtain the previously calculated estimate

$$f'(1) \approx \frac{\Delta f}{\Delta x} = \frac{f(1.001) - f(1)}{1.001 - 1} = \frac{1.001^2 - 1^2}{1.001 - 1} = 2.001.$$

If we examine the average rate of change using the expression containing the variable h , we arrive at the form

$$\begin{aligned} f'(1) \approx \frac{\Delta f}{\Delta x} &= \frac{f(1+h) - f(1)}{(1+h) - 1} = \frac{(1+h)^2 - 1}{h} = \frac{(h^2 + 2h + 1) - 1}{h} \\ &= \frac{h^2 + 2h}{h}. \end{aligned}$$

If we substitute $h = 0$ here, we again end up dividing by zero. However, if we simplify the expression to

$$\frac{\Delta f}{\Delta x} = h + 2$$

and then substitute $h = 0$, we obtain the value 2, which we concluded earlier to be the exact value of the sought instantaneous rate of change. Wonderful! The problem seems to have been resolved.

In fact, it has not. The previous reasoning has a significant flaw. When the expression was simplified using the variable h , we must require that $h \neq 0$, because division by zero is not allowed. Consequently, the simplified form $h + 2$ is not valid when $h = 0$, so the final substitution cannot be made. Why not? Perhaps the most central requirement of all mathematics is that mathematical concepts and results must not lead to contradictions.

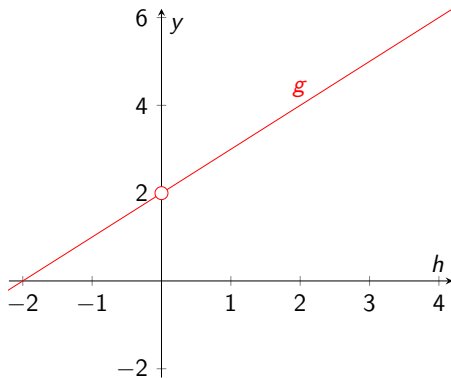
For example, two expressions must not yield two different values through different calculations. If this occurs, nothing derived using these properties can be trusted. Thus, if the value of the expression $(h^2 + h)/h$ is not defined at the point $h = 0$, then the value of the simplified form $2 + h$, $h \neq 0$, is also not defined at that point.

The previous technique does not work. However, there is clearly some sense to it, as it produced the correct value. What is the reason for this? Let us examine the situation through a diagram.

The function

$$g(h) = \frac{h^2 + 2h}{h} = \frac{h}{h}(h + 2)$$

is practically the line $y = h + 2$, which has been extended by the variable h . This extension does not change the line in any way other than making it undefined at the point $h = 0$. The graph of function g is thus a line with a hole at the point $h = 0$, as illustrated in the accompanying image. Ensure that you understand why the variable is h and not x .



The figure shows that the value of the function g approaches the value 2 at the point $h = 0$ without ever reaching it. This missing value would be precisely the correct value for the instantaneous rate of change. However, we cannot refer to it in traditional terms, namely as the value of the function g at the point $h = 0$, because the function is not defined at that point.

What can we do? Well, let's create a new concept that allows us to discuss it. We agree that the function g has a **limit** of 2 at the point $h = 0$, and we denote this by $\lim_{h \rightarrow 0} g(h) = 2$ to distinguish it from the notation for the function value. The limit is, therefore, a value that the function approaches at a certain point from both sides, but which the function may not necessarily attain.

With this new concept, we can state that the instantaneous rate of change of the function f at the point $x = 1$ is the limit

$$f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h},$$

which means that

$$f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{h^2 + 2h}{h} = \lim_{h \rightarrow 0} g(h) = 2.$$

These intermediate steps will be examined in more detail in Chapter 1.1.5

This reasoning can be generalized further to an arbitrary point x of the function $f(x) = x^2$, meaning that its instantaneous rate of change can be obtained at any point as a limit

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

In fact, this limit gives the instantaneous rate of change for any single-variable function at any point where the instantaneous rate of change exists. In other words, it is the definition of the instantaneous rate of change. However, this will be discussed in more detail only in Chapter 1.4, and now we will move on to examine the definition of the limit and the computation of limits more closely.

Exercise

- (a) Deduce, in the manner previously introduced, the rate of change of the function $f(x) = x^3$ at the point $x = 2$. There is no need to pay attention to limits.
- (b) Try to repeat part (a) for the function $f(x) = \cos(x)$ until you notice that it cannot be done. However, determine an approximate value. This illustrates that the concept of limits is not merely a notational quibble, but that a deeper theory of limits is necessary for calculating certain rates of change.

The Rigorous Definition of a Limit

Definition

Let f be a real-valued function whose domain contains points arbitrarily close to a real number $a \in \mathbb{R}$. We say that f has a limit $L \in \mathbb{R}$ at the point a if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon$$

for all x in the domain of f .

(This means that the value of the function $f(x)$ can be made as close as desired to the real number L when the variable x is chosen sufficiently close to the number a .)

In this case, we denote

$$\lim_{x \rightarrow a} f(x) = L$$

or, alternatively,

$$f(x) \rightarrow L, \quad \text{as } x \rightarrow a.$$

In the notation $\lim_{x \rightarrow a} f(x) = L$, the word **lim** comes from the Latin word **limes**, meaning boundary, but it is easy to remember from its English counterpart, limit. The notation $\rightarrow$ is read as “approaches.” The entire expression is read as “the limit of $f(x)$ as x approaches a ” or simply “limit as x approaches a .”

Example

Show that $\lim_{x \rightarrow 2}(2x - 1) = 3$ based on the definition of a limit.

Solution

Let $\varepsilon > 0$ be fixed. The goal is to show that there exists a number $\delta > 0$ (which may depend on ε) such that $|(2x - 1) - 3| < \varepsilon$ whenever $0 < |x - 2| < \delta$.

Sketch:

$$\begin{aligned} |(2x - 1) - 3| < \varepsilon &\Leftrightarrow -\varepsilon < (2x - 1) - 3 < \varepsilon \Leftrightarrow 3 - \varepsilon < 2x - 1 < 3 + \varepsilon \\ &\Leftrightarrow 2 - \frac{\varepsilon}{2} < x < 2 + \frac{\varepsilon}{2} \Leftrightarrow -\frac{\varepsilon}{2} < x - 2 < \frac{\varepsilon}{2} \Leftrightarrow |x - 2| < \frac{\varepsilon}{2} =: \delta. \end{aligned}$$

Set $\delta = \frac{\varepsilon}{2}$ and choose $x \in \mathbb{R} \setminus \{2\}$ such that $|x - 2| < \delta = \frac{\varepsilon}{2}$. Now it follows that

$$\begin{aligned} |x - 2| < \frac{\varepsilon}{2} &\Rightarrow 2 - \frac{\varepsilon}{2} < x < 2 + \frac{\varepsilon}{2} \Rightarrow 4 - \varepsilon < 2x < 4 + \varepsilon \\ &\Rightarrow -\varepsilon < 2x - 4 < \varepsilon \Rightarrow |(2x - 1) - 3| < \varepsilon. \end{aligned}$$

Hence $\lim_{x \rightarrow 2}(2x - 1) = 3$.

In fact, as we shall see, a function $f: \mathbb{R} \rightarrow \mathbb{R}$ is **continuous** at a point $a \in \mathbb{R}$ precisely when there holds $\lim_{x \rightarrow a} f(x) = f(a)$, i.e., **the limit of a continuous function coincides with function evaluation**. The situation is more interesting if this property cannot be exploited.

Example

Show that $\lim_{x \rightarrow 0} x \sin(\frac{1}{x}) = 0$ based on the definition of a limit.

Solution

Let $\varepsilon > 0$ be fixed. The goal is to show that there exists a number $\delta > 0$ (which may depend on ε) such that $|x \sin(\frac{1}{x})| < \varepsilon$ whenever $0 < |x| < \delta$.

Sketch:

$$|x \sin(\frac{1}{x})| < \varepsilon \quad \stackrel{|\sin(\frac{1}{x})| \leq 1}{\Leftrightarrow} \quad |x \sin(\frac{1}{x})| \leq |x| < \varepsilon =: \delta.$$

Choose $\delta = \varepsilon$ and let $x \in \mathbb{R} \setminus \{0\}$. Now

$$|x| < \varepsilon \quad \stackrel{|\sin(\frac{1}{x})| \leq 1}{\Rightarrow} \quad \left| x \sin \left(\frac{1}{x} \right) \right| \leq |x| < \varepsilon.$$

Hence $\lim_{x \rightarrow 0} x \sin(\frac{1}{x}) = 0$.

Example

Show that $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1} = 2$ based on the definition of a limit.

Solution

Let $\varepsilon > 0$ be fixed. The goal is to show that there exists a number $\delta > 0$ (which may depend on ε) such that $\left| \frac{x^2-1}{x-1} \right| < \varepsilon$ whenever $0 < |x-1| < \delta$. Importantly, since $x \neq 1$, we can write $\frac{x^2-1}{x-1} = x+1$.

Sketch:

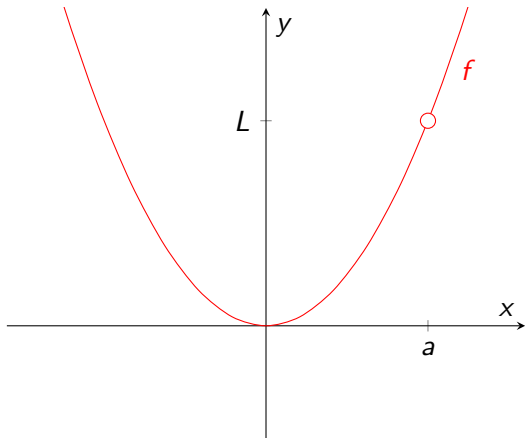
$$\left| \frac{x^2-1}{x-1} - 2 \right| < \varepsilon \stackrel{x \neq 1}{\Leftrightarrow} |(x+1) - 2| < \varepsilon \Leftrightarrow |x-1| < \varepsilon =: \delta$$

Set $\delta = \varepsilon$ and choose $x \in \mathbb{R} \setminus \{1\}$ such that $|x-1| < \delta = \varepsilon$. Now it follows that

$$|x-1| < \varepsilon \quad \Rightarrow \quad |(x+1) - 2| < \varepsilon \quad \Rightarrow \quad \left| \frac{x^2-1}{x-1} - 2 \right| < \varepsilon.$$

Hence $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1} = 2$.

The definition is straightforward to understand through the graph of the function: the limit at the point $x = a$ is L if the graph approaches the height L when approaching the point $x = a$ from either side.



At this stage, the limit of a function can only be roughly deduced based on the simplified definition either from the graph or by testing individual values. For example, we can examine the limit of the function

$f(x) = \frac{x}{\sin(x)}$ at the point $x = 0$ as follows:

$$f(0.1) \approx 1.002$$

$$f(-0.1) \approx 1.002$$

$$f(0.01) \approx 1.00002$$

$$f(-0.01) \approx 1.00002$$

$$f(0.001) \approx 1.0000002$$

$$f(-0.001) \approx 1.0000002.$$

Based on these calculations, it strongly appears that $\lim_{x \rightarrow 0} f(x) = 1$. Ensure that you also see this from the graph.

Basic Exercise

Determine the limits of the following functions at the point $x = 0$ by testing small values:

$$(a) f(x) = \frac{\sin(x)}{x}$$

$$(b) g(x) = \cos\left(\frac{1}{x}\right).$$

Let's also pay attention to a couple of points in the definition that are not entirely obvious. First, the fact that L is a real number implies that the limit must be finite. Later, we will also examine limits that may be infinitely large or small, which are referred to as infinite limits in contrast to ordinary limits. Sometimes limits may be specifically referred to as finite limits, but in practice, finiteness is an unnecessary addition, as non-infinite limits are by default finite. Secondly, it is essential to understand that the value of the function at the point under consideration does not matter for the existence of the limit, as illustrated in the following example.

Example

Determine the limits of the following functions at the point $x = 1$ based on their graphs:

$$(a) f_1(x) = x + 1$$

$$(b) f_2(x) = \frac{(x-1)(x+1)}{x-1}$$

$$(c) f_3(x) = \begin{cases} x+1, & x \neq 1 \\ 4, & x = 1 \end{cases}$$

$$(d) f_4(x) = \begin{cases} x+1, & x < 1 \\ x+2, & x \geq 1 \end{cases}$$

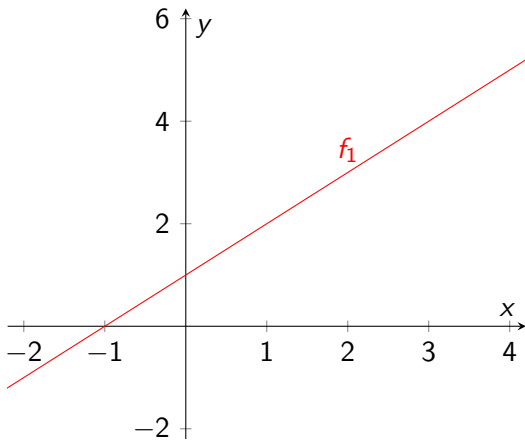
$$(e) f_5(x) = \frac{1}{x-1}$$

$$(f) f_6(x) = \frac{1}{(x-1)^2}$$

$$(g) f_7(x) = \sin\left(\frac{1}{x-1}\right).$$

Solution

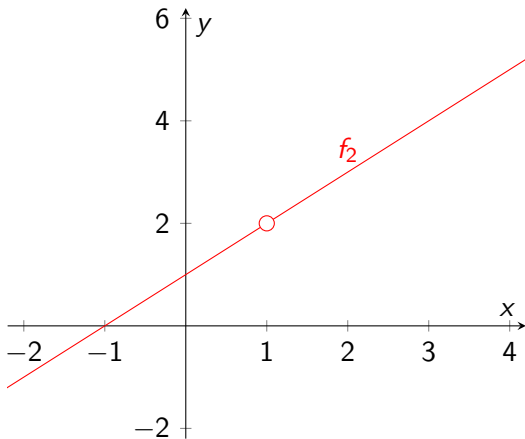
(a)



The graph of the function approaches the value 2 at the point $x = 1$ from either side, so $\lim_{x \rightarrow 1} f_1(x) = 2$.

Solution (continued)

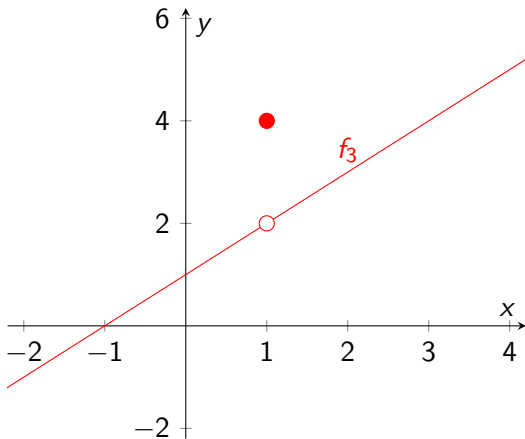
(b)



The function is otherwise the same as in part (a), but it is not defined at the point $x = 1$, meaning that its graph has a hole at this point. Nevertheless, the graph of the function approaches the value 2 at that point from either side, so $\lim_{x \rightarrow 1} f_2(x) = 2$.

Solution (continued)

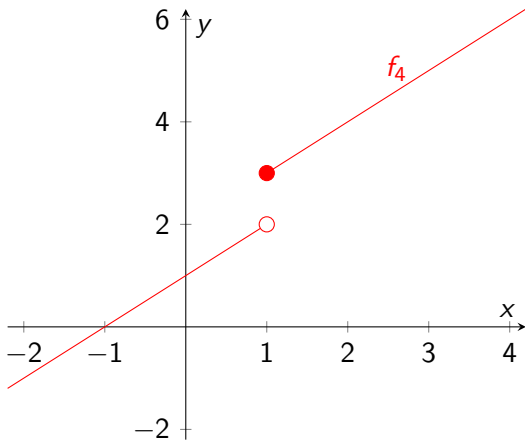
(c)



The function is otherwise the same as in the previous parts, but its value at the point $x = 1$ is 4. This does not change the situation regarding the limit in any way, and it still holds that $\lim_{x \rightarrow 1} f_3(x) = 2$.

Solution (continued)

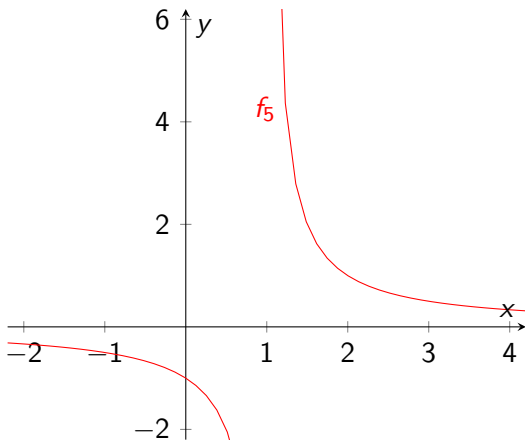
(d)



Now the value of the function approaches 2 as the point $x = 1$ is approached from the left, and 3 as that point is approached from the right. Thus, the function f_4 does not have a limit at the point $x = 1$.

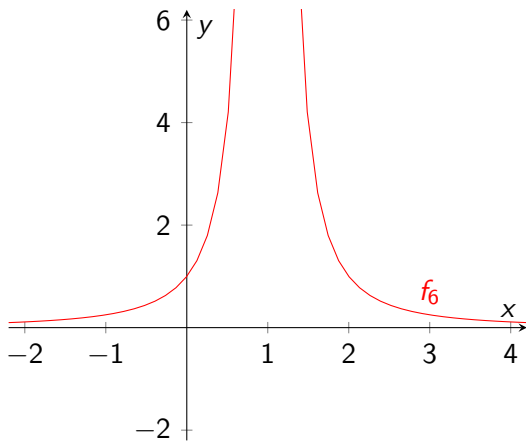
Solution (continued)

(e)



In this case, there are two problems. First, the value of the function does not approach the same value from both sides at $x = 1$. On the other hand, it does not approach a finite number from either side. Therefore, the limit $\lim_{x \rightarrow 1} f_5(x)$ does not exist.

Solution (continued)
(f)



In this case, the value of the function approaches infinity from both sides at $x = 1$, but since the limit must be finite, the limit $\lim_{x \rightarrow 1} f_6(x)$ does not exist. (However, in this case, we can say that the function has an **infinite** limit at $x = 1$, which is ∞ . This will be discussed in a later chapter.)

Solution (continued)

- (g) This is clearest when looking at a graphing program where you can zoom in. The function in question oscillates more and more as we approach $x = 1$ and therefore does not approach any specific value. Thus, the limit $\lim_{x \rightarrow 1} f_7(x)$ does not exist.

In the previous example, we observed cases (d) through (g) where the limit does not exist. Next, we will summarize such cases.

Theorem

A function does not have a limit at $x = a$ if

- the function value approaches different values from either side as we approach $x = a$
- the function value approaches infinity or negative infinity from either side or both sides as we approach $x = a$
- the function does not approach any specific value
- the function is not defined on one side of $x = a$.

Examples of the first three cases were seen in the previous task. The last case can be illustrated by the function $f: [1, 2] \rightarrow \mathbb{R}$, $f(x) = x + 5$ at points $x = 1$ and $x = 2$.

Basic Exercise

Determine the limit of the following functions at $x = 0$ based on their graphs, if such a limit exists:

$$(a) f_1: \mathbb{R}_{\neq 0} \rightarrow \mathbb{R}, \quad f_1(x) = \frac{x^3}{x} + 1 \quad (b) f_2: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}, \quad f_2(x) = \sqrt{x}$$

$$(c) f_3: \mathbb{R} \rightarrow \mathbb{R}, \quad (d) f_4: \mathbb{R} \rightarrow \mathbb{R}, \quad f_4(x) = e^x$$

$$f_3(x) = \begin{cases} x^2 + 1, & x < 1 \\ x, & x \geq 1 \end{cases}$$

$$(e) f_5: \mathbb{R}_{\neq 0} \rightarrow \mathbb{R}, \quad f_5(x) = \frac{x}{x^4} \quad (f) f_6: \mathbb{R} \rightarrow \mathbb{R},$$
$$f_6(x) = \begin{cases} x^2, & x \neq 0 \\ 2021, & x = 0 \end{cases}$$

$$(g) f_7: \{x \in \mathbb{R} \mid x \neq n\pi, n \in \mathbb{Z}\} \rightarrow \mathbb{R}, \quad f_7(x) = \tan\left(\frac{1}{x}\right).$$

One-Sided Limits

If the limit does not exist but the function approaches some value from one side or both sides at the point in question, we can refer to these values as one-sided limits.

Definition

Let f be a real-valued function defined over the interval $(a - h, a)$ with $a \in \mathbb{R}$ and $h > 0$. Then f has a left-hand limit $L \in \mathbb{R}$ at $x = a$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$a - \delta < x < a \implies |f(x) - L| < \varepsilon.$$

(This means that the function $f(x)$ can be made as close as desired to a real number L by choosing x sufficiently close to a from the left side.)

In this case, we denote $\lim_{x \rightarrow a^-} f(x) = L$ or $f(x) \rightarrow L$, as $x \rightarrow a^-$.

Definition

Let f be a real-valued function defined over the interval $(a, a + h)$ with $a \in \mathbb{R}$ and $h > 0$. Then f has a right-hand limit $L \in \mathbb{R}$ at $x = a$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$a < x < a + \delta \implies |f(x) - L| < \varepsilon.$$

(This means that the value of the function $f(x)$ can be made as close as desired to a real number L by choosing x sufficiently close to a from the right side.)

In this case, we denote $\lim_{x \rightarrow a^+} f(x) = L$ or $f(x) \rightarrow L$, as $x \rightarrow a^+$.

Example

Show that $\lim_{x \rightarrow 0^+} \sqrt{x} = 0$ based on the definition of a one-sided limit.

Solution

Let $\varepsilon > 0$ be fixed. The goal is to show that there exists a number $\delta > 0$ (which may depend on ε) such that $|\sqrt{x}| < \varepsilon$ whenever $0 < x < \delta$.

Sketch:

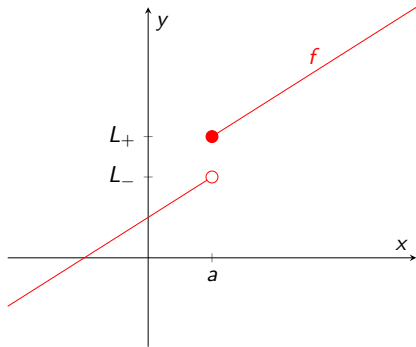
$$|\sqrt{x}| < \varepsilon \stackrel{x \geq 0}{\iff} 0 < x < \varepsilon^2 =: \delta.$$

Set $\delta = \varepsilon^2$ and $x > 0$. Then

$$0 < |x| < \delta = \varepsilon^2 \quad \Rightarrow \quad |\sqrt{x}| < \varepsilon.$$

Hence $\lim_{x \rightarrow 0^+} \sqrt{x} = 0$.

One-sided limits are illustrated in the accompanying figure, which uses the notations $L_- = \lim_{x \rightarrow a^-} f(x)$ and $L_+ = \lim_{x \rightarrow a^+} f(x)$.



The existence of a limit can be illustrated using one-sided limits.

Theorem

A function has a limit at $x = a$ if and only if its one-sided limits are equal. Formally:

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L.$$

Example

Based on the graphs, determine the one-sided limits at $x = 1$ for the functions from the previous example:

$$(a) f_1(x) = x + 1$$

$$(b) f_2(x) = \frac{(x-1)(x+1)}{x-1}$$

$$(c) f_3(x) = \begin{cases} x + 1, & x \neq 1 \\ 4, & x = 1 \end{cases}$$

$$(d) f_4(x) = \begin{cases} x + 1, & x < 1 \\ x + 2, & x \geq 1 \end{cases}.$$

Based on the one-sided limits, determine whether the function has a limit at that point.

Solution

$$(a) \lim_{x \rightarrow 1^-} f_1(x) = 2 \text{ and } \lim_{x \rightarrow 1^+} f_1(x) = 2, \text{ therefore } \lim_{x \rightarrow 1} f_1(x) = 2.$$

$$(b) \lim_{x \rightarrow 1^-} f_2(x) = 2 \text{ and } \lim_{x \rightarrow 1^+} f_2(x) = 2, \text{ therefore } \lim_{x \rightarrow 1} f_2(x) = 2.$$

$$(c) \lim_{x \rightarrow 1^-} f_3(x) = 2 \text{ and } \lim_{x \rightarrow 1^+} f_3(x) = 2, \text{ therefore } \lim_{x \rightarrow 1} f_3(x) = 2.$$

$$(d) \lim_{x \rightarrow 1^-} f_4(x) = 2 \text{ and } \lim_{x \rightarrow 1^+} f_4(x) = 3, \text{ therefore the function } f_4 \text{ does not have a limit at } x = 1.$$

Basic Exercise

Determine based on the graphs the one-sided limits at $x = 0$ for the functions:

$$(a) f_1: \mathbb{R}_{\neq 0} \rightarrow \mathbb{R}, \quad f_1(x) = \frac{x^3}{x} + 1 \quad (b) f_2: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}, \quad f_2(x) = \sqrt{x}$$

$$(c) f_3: \mathbb{R} \rightarrow \mathbb{R}, \quad (d) f_4: \mathbb{R} \rightarrow \mathbb{R}, \quad f_4(x) = e^x$$

$$f_3(x) = \begin{cases} x^2 + 1, & x < 1 \\ x, & x \geq 1 \end{cases}$$

$$(e) f_5: \mathbb{R}_{\neq 0} \rightarrow \mathbb{R}, \quad f_5(x) = \frac{x}{x^4} \quad (f) f_6: \mathbb{R} \rightarrow \mathbb{R},$$
$$f_6(x) = \begin{cases} x^2, & x \neq 0 \\ 2021, & x = 0 \end{cases}$$

$$(g) f_7: \{x \in \mathbb{R} \mid x \neq n\pi, n \in \mathbb{Z}\} \rightarrow \mathbb{R}, \quad f_7(x) = \tan\left(\frac{1}{x}\right).$$

Based on the one-sided limits at $x = 0$, determine whether the function has a limit at that point.

Continuity

At this stage, we will briefly discuss the continuity of a function, as it is needed to justify intermediate steps in calculations related to limits. Continuity will be covered in more detail later in Chapter 1.2.

The idea of continuity of a function is quite clear based on its graph: a function is continuous at a given point if its graph does not break. This can be formulated precisely using one-sided limits.

Definition

A function f is continuous at $x = a$ if and only if the following holds:

$$\lim_{x \rightarrow a^-} f(x) = f(a) = \lim_{x \rightarrow a^+} f(x).$$

Thus, a function is continuous at the point in question if and only if its value and the one-sided limits are the same at that point.

The equality of the one-sided limits implies the existence of the limit at $x = a$, so the previous condition can be summarized as:

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Therefore, a function is continuous at $x = a$ if and only if its value and limit are the same at that point.

Example

Which of the functions

(a) $f_1(x) = x + 1$

(b) $f_2(x) = \frac{(x-1)(x+1)}{x-1}$

(c) $f_3(x) = \begin{cases} x+1, & x \neq 1 \\ 4, & x = 1 \end{cases}$

(d) $f_4(x) = \begin{cases} x+1, & x < 1 \\ x+2, & x \geq 1 \end{cases}.$

are continuous at $x = 1$ based on the previous condition?

Solution

- (a) The function f_1 is continuous at $x = 1$ since (based on the graph) we have

$$\lim_{x \rightarrow 1^-} f_1(x) = 2, \quad f_1(1) = 2 \quad \text{and} \quad \lim_{x \rightarrow 1^+} f_1(x) = 2.$$

- (b) The function f_2 is not continuous at $x = 1$ since the value $f_2(1)$ does not exist, so the condition for continuity cannot be satisfied.

- (c) The function f_3 is not continuous at $x = 1$ since (based on the graph) we have

$$\lim_{x \rightarrow 1} f_3(x) = 2 \quad \text{and} \quad f_3(1) = 4,$$

so $\lim_{x \rightarrow 1} f(x) \neq f(1)$.

- (d) The function f_4 is not continuous at $x = 1$ since (based on the graph) we have

$$\lim_{x \rightarrow 1^-} f_4(x) = 2 \quad \text{and} \quad f_4(1) = 3,$$

so $\lim_{x \rightarrow 1^-} f(x) \neq f(1)$.

Basic Exercise

Which of the functions

$$(a) f_1: \mathbb{R}_{\neq 0} \rightarrow \mathbb{R}, \quad f_1(x) = \frac{x^3}{x} + 1 \quad (b) f_2: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}, \quad f_2(x) = \sqrt{x}$$

$$(c) f_3: \mathbb{R} \rightarrow \mathbb{R}, \quad (d) f_4: \mathbb{R} \rightarrow \mathbb{R}, \quad f_4(x) = e^x$$

$$f_3(x) = \begin{cases} x^2 + 1, & x < 1 \\ x, & x \geq 1 \end{cases}$$

$$(e) f_5: \mathbb{R}_{\neq 0} \rightarrow \mathbb{R}, \quad f_5(x) = \frac{x}{x^4} \quad (f) f_6: \mathbb{R} \rightarrow \mathbb{R},$$
$$f_6(x) = \begin{cases} x^2, & x \neq 0 \\ 2021, & x = 0 \end{cases}$$

$$(g) f_7: \{x \in \mathbb{R} \mid x \neq n\pi, n \in \mathbb{Z}\} \rightarrow \mathbb{R}, \quad f_7(x) = \tan\left(\frac{1}{x}\right).$$

are continuous at $x = 0$?

Calculating Limits

First, we will go through how to calculate limits in practice for rational functions, and only after that will we justify the basis for the intermediate steps in the calculations.

Example

Determine the following limits, or show that they do not exist:

$$(a) \lim_{x \rightarrow 1} (x + 1) \qquad (b) \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} \qquad (c) \lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 2x}$$

$$(d) \lim_{x \rightarrow 0} \frac{x^3}{x^2} \qquad (e) \lim_{x \rightarrow 1} f(x),$$

$$f(x) = \begin{cases} x + 1, & x < 1 \\ \frac{(x - 1)(x + 2)}{x - 1}, & x > 1 \end{cases}$$

$$(f) \lim_{x \rightarrow 1} g(x), \quad g(x) = \begin{cases} x + 1, & x \neq 1 \\ 4, & x = 1 \end{cases}.$$

Solution

- (a) If the function under consideration is not piecewise defined and direct substitution $x = a$ results in a numerical value when calculating the limit $\lim_{x \rightarrow a} f(x)$, then the limit can be obtained simply this way. This is the case here:

$$\lim_{x \rightarrow 1} (x + 1) = 1 + 1 = 2.$$

- (b) If direct substitution when calculating the limit $\lim_{x \rightarrow a} f(x)$ leads to an undefined expression[†] “0/0”, then the function must be simplified by the expression $x - a$ and direct substitution should be tried again afterward. Now we have

$$\lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} = \frac{\text{“0”}}{0},$$

so the function should be simplified by the expression $x - 1$ and then the substitution $x = 1$ should be tried. This now succeeds easily since the function is already in a nice form:

$$\lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} = \lim_{x \rightarrow 1} (x + 1) = 1 + 1 = 2.$$

[†]The expression “0/0” has quotation marks because it is not a meaningful expression. It does not signify anything.

Solution (continued)

(c) Direct substitution leads again to the expression “0/0”:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 2x} = \frac{“0”}{0}.$$

Thus, the function must be simplified by the expression $x - 2$, meaning the numerator and denominator need to be expressed as products that include the factor $x - 2$. Later, we will justify why this always works for rational functions when direct substitution leads to “0/0”. Manipulating the denominator can be done straightforwardly: $x^2 - 2x = x(x - 2)$. The numerator can also be immediately seen as the product of two binomials, but we will process it in a universally applicable way that is based on the fact that polynomials can be expressed as products of their roots. So, let us solve for the roots of the numerator:

$$x^2 - 4 = 0 \iff x^2 = 4 \iff x = \pm 2.$$

Solution (continued)

Thus, the numerator can be written as

$$x^2 - 4 = (x - (-2))(x - 2) = (x + 2)(x - 2).$$

Now let's calculate the limit after simplification:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 2x} = \lim_{x \rightarrow 2} \frac{(x + 2)(x - 2)}{x(x - 2)} = \lim_{x \rightarrow 2} \frac{x + 2}{x} = \frac{2 + 2}{2} = 2.$$

It worked.

- (d) If one simplification is not enough, it should be repeated until a finite number is obtained as the limit:

$$\lim_{x \rightarrow 0} \frac{x^3}{x^2} = \frac{"0"}{0},$$

$$\lim_{x \rightarrow 0} \frac{x^3}{x^2} = \lim_{x \rightarrow 0} \frac{x^2}{x} = \frac{"0"}{0},$$

$$\lim_{x \rightarrow 0} \frac{x^3}{x^2} = \lim_{x \rightarrow 0} x = 0.$$

Solution (continued)

- (e) If the function's definition changes at the point under consideration, both one-sided limits should be calculated. These can be obtained like the previous limits, either by direct substitution or by substitution after simplification if necessary. The crucial part is just to substitute into the correct expressions. Now, the case $x \rightarrow 1^-$ corresponds to the expression for the case $x < 1$, meaning we have

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x + 1) = \lim_{x \rightarrow 1} (x + 1) = 1 + 1 = 2.$$

The case $x \rightarrow 1^+$ corresponds to the expression for $x \geq 1$, meaning we have

$$\begin{aligned} \lim_{x \rightarrow 1^+} f(x) &= \lim_{x \rightarrow 1^+} \frac{(x-1)(x+2)}{x-1} = \lim_{x \rightarrow 1} \frac{(x-1)(x+2)}{x-1} = \lim_{x \rightarrow 1} (x+2) \\ &= 1 + 2 = 3. \end{aligned}$$

Since $\lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$, the limit we are looking for does not exist.

Solution (continued)

- (f) As in the previous case, both one-sided limits need to be calculated here. However, since the expression is the same on both sides, only one calculation is sufficient:

$$\lim_{x \rightarrow 1^-} g(x) = \lim_{x \rightarrow 1^+} g(x) = \lim_{x \rightarrow 1} g(x) = \lim_{x \rightarrow 1} (x + 1) = 1 + 1 = 2.$$

Let it be noted at the end that not all calculations need to be explicitly shown, but in this example, even errors have been left visible for illustration purposes. For example, in part (c), one could write directly:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 2x} = \lim_{x \rightarrow 2} \frac{(x + 2)(x - 2)}{x(x - 2)} = \lim_{x \rightarrow 2} \frac{x + 2}{x} = \frac{2 + 2}{2} = 2.$$

Example

If possible, determine the limit

$$\lim_{x \rightarrow 2} \frac{|x - 2|}{x^2 + x - 6}.$$

Solution

The expression $|x - 2|$ can take two forms depending on whether $x < 2$ or $x \geq 2$, so we need to consider the one-sided limits separately:

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2} \overbrace{\frac{-(x-2)}{x^2 + x - 6}}^{\frac{0}{0}} = \lim_{x \rightarrow 2} \frac{-(x-2)}{(x-2)(x+3)} = \lim_{x \rightarrow 2} \frac{-1}{x+3} = \frac{-1}{2+3} = -\frac{1}{5}$$

and

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2} \overbrace{\frac{x-2}{x^2 + x - 6}}^{\frac{0}{0}} = \lim_{x \rightarrow 2} \frac{(x-2)}{(x-2)(x+3)} = \lim_{x \rightarrow 2} \frac{1}{x+3} = \frac{1}{5},$$

so

$$\lim_{x \rightarrow 2^+} f(x) \neq \lim_{x \rightarrow 2^-} f(x),$$

and therefore, the limit at $x = 2$ does not exist.

Basic Exercise

Calculate the limit $\lim_{x \rightarrow 3} f(x)$ in the following cases or show that it does not exist:

$$(a) f(x) = \frac{x^2 - 9}{x + 3}$$

$$(b) f(x) = \frac{x^2 - 9}{x - 3}$$

$$(c) f(x) = \begin{cases} -x + 3, & x < 3 \\ x^2 - 9 & x \geq 3 \end{cases}$$

$$(d) f(x) = \begin{cases} -x + 3, & x < 0 \\ x^2 - 9 & x \geq 0 \end{cases}$$

$$(e) f(x) = \begin{cases} -x + 3, & x < 3 \\ x^2 - 4 & x \geq 3 \end{cases}$$

$$(f) f(x) = \begin{cases} -x + 3, & x = 3 \\ x^2 - 4 & x \neq 3 \end{cases}$$

Exercise

Calculate the limit

$$\lim_{x \rightarrow 3} \frac{x^3 - 7x^2 + 15x - 9}{2x^2 - 12x + 18}.$$

Calculating limits of rational functions is therefore quite straightforward. But what do these calculations really depend on? If we write the expression for (a) symbolically, it becomes

$$\lim_{x \rightarrow 1} f(x) = f(1) = 2.$$

There should be something familiar here. The first equality concerns continuity. When a function is continuous at $x = a$, then by the property of continuity we have $\lim_{x \rightarrow a} f(x) = f(a)$, meaning the limit can be calculated at the point a through direct substitution.

What does it mean to end up with the expression “0/0” in cases (b) to (d)? It indicates that the function is not defined at that point, implying that the function is not continuous at this point. Therefore, direct substitution does not work. How do we know that we can simplify the numerator and denominator by the expression $x - a$?

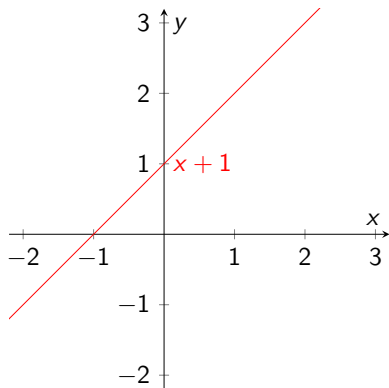
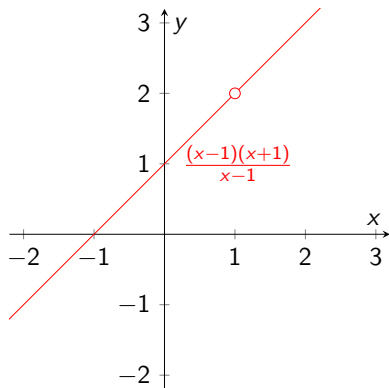
In a rational function, both the numerator and denominator contain some polynomial. The fact that the polynomial equals zero at the substitution point $x = a$ means that the substituted value a is a root of the polynomial. Consequently, the polynomial can be expressed in a form that includes the factor $x - a$, because polynomials can be written as products of their roots. Thus, in the case of the expression “0/0”, the numerator and denominator of the rational function will always inevitably contain the factors $x - a$, which can be cancelled out.

But why does this simplification help in calculating the limit? For example, when the expression $x + 1$ is multiplied by a non-zero real number, its value does not change for any value of the variable x . When it is multiplied by the expression $x - 1$, its value does not change for any other value of the variable x , except in the case where $x = 1$, at which point the expression becomes undefined. Conversely, in the simplification

$$\frac{(x - 1)(x + 1)}{x - 1} = x + 1$$

the original expression only changes in that it becomes defined at $x = 1$.

From the perspective of graphs, this means that the functions have the same graphs everywhere except that one has a hole at $x = 1$. When this hole is removed, the function becomes continuous, as illustrated in the accompanying diagrams.



Thus, the limit $\lim_{x \rightarrow 1} (x + 1)$ is obtained through direct substitution based on continuity, but the limit $\lim_{x \rightarrow 1} \left(\frac{(x-1)(x+1)}{x-1} \right)$ cannot be obtained.

But wasn't it stated in the introduction that one should not write

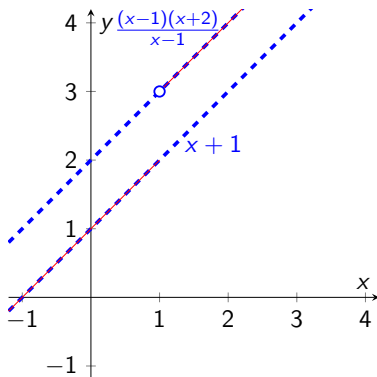
$$\frac{(x-1)(x+1)}{x-1} = x+1$$

and then consider the situation at $x = 1$, because the expressions are not the same at that point? It was stated, but now we instead write

$$\lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{x-1} = \lim_{x \rightarrow 1} (x+1),$$

which means we do not equate the entire expressions but only their limits at $x = 1$. According to the discussion above, these limits are the same since the graphs of the expressions differ only in that the former has a hole at $x = 1$ and the latter does not.

In part (e), the value that the expression $g_-(x) = x + 1$ approaches as $x \rightarrow 1^-$ is exactly the same value that the expression would take at $x = 1$, so this value can be obtained by direct substitution: $g_-(1)$. On the other hand, the expression $g_+(x) = \frac{(x-1)(x+2)}{x-1}$ is not defined at that point, so the value it approaches as $x \rightarrow 1^+$ is the same as the expression's limit $\lim_{x \rightarrow 1} g_+(x)$ at that point. This is illustrated in the accompanying figure, where the red graph corresponds to the function in part (e), and the blue graphs (which are drawn as dashed lines to show the red graph) represent what g_- and g_+ would look like in the largest possible domain of definition.



More generally: The calculation of one-sided limits reduces to the calculation of ordinary limits. For example, in the case of the function

$$f(x) = \begin{cases} f_-(x), & x < a \\ f_+(x), & x > a \end{cases}$$

the one-sided limits satisfy $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a} f_-(x)$ and $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a} f_+(x)$. If f_- or f_+ is continuous at the point in question, it simplifies to $\lim_{x \rightarrow a^-} f(x) = f_-(a)$ or $\lim_{x \rightarrow a^+} f(x) = f_+(a)$.

The reasoning for part (f) can be justified like that of part (e), but the following explanation is likely clearer. The function differs from the function $f(x) = x + 1$ only at the point $x = 1$. Therefore, they have the same limit at every point, and thus particularly at the point $x = 1$, so the limit can be examined using the function f .

Now let's return for a moment to what we initially started discussing regarding limits. In the introduction, it was concluded that the instantaneous rate of change of the function $f(x) = x^2$ at the point $x = 1$ could be calculated precisely as follows:

$$f'(1) = \frac{f(1+h) - f(1)}{(1+h) - 1} = \frac{(1+h)^2 - 1}{h} = \frac{h^2 + 2h}{h} \quad [h \neq 0] \quad (1) = h + 2 \quad [h = 0] = 2$$

but that this cannot be done, because the equality (1) does not hold at $h = 0$, and therefore the substitution $h = 0$ is not allowed. The limit circumvents this problem, since one can write

$$f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{(1+h) - 1} = \lim_{h \rightarrow 0} \frac{h^2 + 2h}{h} \stackrel{(1)}{=} \lim_{h \rightarrow 0} (h + 2) \stackrel{(2)}{=} 0 + 2 = 2.$$

Now, the equality (1) holds at $h = 0$, because we are not comparing the function values but the limits, which are the same. The equality (2), on the other hand, relies on continuity, which is based on the fact that in equality (1), we transitioned from a discontinuous function at $h = 0$ to a function that is continuous at that point. Thus, the limit enables the precise calculation of the function's rate of change. This is the original purpose of the limit.

Exercise

Determine the instantaneous rate of the function $f(x) = x^2$ at an arbitrary point x .

The techniques discussed above primarily work only in the case of rational functions. For example, when attempting to calculate the instantaneous rate of the function $f(x) = \sin(x)$ at an arbitrary point x in the same manner, one ends up in the situation

$$f'(1) = \lim_{h \rightarrow 0} \frac{\sin(1+h) - \sin(1)}{h} = \frac{0}{0},$$

which cannot be resolved by suitable cancellation. The calculation of this limit is successful with a technique called the **squeeze theorem**. This technique will be a deeper topic in this course and will be addressed a bit later in Chapter 1.1.8. After the discussion of non-essential limits, another general limit calculation technique known as **L'Hôpital's rule** will also be introduced in Chapter 1.1.11.

Let's go through another case where cancellation works. This concerns functions that involve square roots.

Example

Determine the limit

$$\lim_{x \rightarrow 4} \frac{x^2 - 16}{\sqrt{x} - 2}.$$

Solution

Direct substitution leads to problems:

$$\lim_{x \rightarrow 4} \frac{x^2 - 16}{\sqrt{x} - 2} = \frac{0}{0}.$$

This situation can be resolved as with rational functions by canceling $x - 4$. The expression can be obtained in the denominator by rationalizing with $\sqrt{x} + 2$:

$$\begin{aligned} \lim_{x \rightarrow 4} \frac{x^2 - 16}{\sqrt{x} - 2} &= \lim_{x \rightarrow 4} \frac{(x^2 - 16)(\sqrt{x} + 2)}{(\sqrt{x} - 2)(\sqrt{x} + 2)} = \lim_{x \rightarrow 4} \frac{(x + 4)(x - 4)(\sqrt{x} + 2)}{x - 4} \\ &= \lim_{x \rightarrow 4} (x + 4)(\sqrt{x} + 2) = (4 + 4)(\sqrt{4} + 2) = 32. \end{aligned}$$

The same technique used to remove the square root from the denominator in the previous example is also applied in division of complex numbers by rationalizing them into a form where the divisor is real. This topic will arise later in the course Differential Equations when discussing complex numbers.

Basic Exercise

Calculate the limit of the function

$$f(x) = \frac{2x - 6}{\sqrt{3} - \sqrt{x}}$$

at the point $x = 3$ or show that it does not exist.

Exercise

Come up with a function (a) that is a rational function (b) a function containing a square root in the denominator, such that it has a limit at the point $x = -2$, but calculating the limit by direct substitution leads to the expression “0/0”.

Rules for Calculating Limits

In the previous chapter, limits were calculated using continuity. We can also derive rules for calculating limits that can sometimes be useful in dealing with limits. On the other hand, using these rules, limits can be calculated without appealing to continuity. While this is not practically significant, it is important for the logical coherence of the theory, which will be discussed in the next chapter.

The most typical rules for calculating the limit of a function are:

Theorem

Let $k \in \mathbb{R}$, and let the functions f and g have limits at $x = a$. Then the following holds:

$$(1) \lim_{x \rightarrow a} k = k$$

$$(2) \lim_{x \rightarrow a} (kf(x)) = k \lim_{x \rightarrow a} f(x)$$

$$(3) \lim_{x \rightarrow a} (f(x) \pm g(x)) = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$$

$$(4) \lim_{x \rightarrow a} (f(x)g(x)) = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

$$(5) \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}, \quad \lim_{x \rightarrow a} g(x) \neq 0.$$

In these formulas, it is assumed that the limits $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist.

When the obvious result

$$\lim_{x \rightarrow a} x = a, \quad (1)$$

is added to the previous formulas, we obtain sufficient formulas to calculate the limits of polynomials and rational functions without appealing to continuity.

Example

In an earlier Example, we determined the limit $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 2x^2}$ using continuity. Let's repeat this using the previous rules for calculating limits:

$$\lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{x(x-2)} \stackrel{(*)}{=} \lim_{x \rightarrow 2} \frac{x+2}{x} = \frac{\lim_{x \rightarrow 2} (x+2)}{\lim_{x \rightarrow 2} x} = \frac{\lim_{x \rightarrow 2} x + \lim_{x \rightarrow 2} 2}{\lim_{x \rightarrow 2} x} = \frac{\lim_{x \rightarrow 2} x + 2}{\lim_{x \rightarrow 2} x} \stackrel{(*)}{=} \frac{2+2}{2} = 2.$$

Equality $(*)$ is again based on the fact that although the functions do not have the same value at $x = 2$, they have the same limit at that point. Note that equality $(*)$ can be justified by property (1) instead of continuity.

The previous example illustrates that calculating limits based on continuity is often more straightforward than using formulas. Next, let's consider a problem where the rules for calculating limits are genuinely useful.

Exercise

Introduction: In the previous chapter, it was noted that the instantaneous rate of change of the function $f(x) = \sin(x)$ is described by the limit

$$\lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin(x)}{h}$$

but that this limit cannot be calculated using the same technique as for the limits of rational functions. However, using a bit of trigonometry and the rules for calculating limits, it can be shown that

$$\lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin(x)}{h} = \sin(x) \lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} + \cos(x) \lim_{h \rightarrow 0} \frac{\sin(h)}{h}. \quad (2)$$

This means that computing the original limit is possible if we can determine

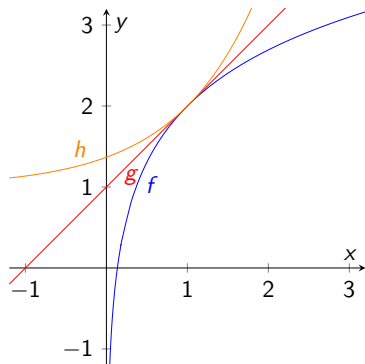
$$\lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} \quad \text{and} \quad \lim_{h \rightarrow 0} \frac{\sin(h)}{h}.$$

This will be done later in Chapter 1.1.8 using the squeeze theorem.

Task: Prove the equation (2) true.

Squeeze Theorem

The techniques based on simplification or formulas discussed earlier do not work in all cases, as observed in the previous Example. The treatment of some interesting problem cases can be successfully handled using a technique called the Squeeze Theorem, which will be introduced next. If for every value of the variable x , the inequality $f(x) \leq g(x) \leq h(x)$ holds and additionally $f(1) = h(1) = 2$, then it must also hold that $g(1) = 2$. This is illustrated in the accompanying figure.



In a similar manner, it is possible to deduce more generally the limit of the function g at the point $x = a$ using a suitable lower bounding function f and an upper bounding function h , which restrict the limit of the function g to a specific value at the point of interest.

Theorem

Let $f(x) \leq g(x) \leq h(x)$ hold for every $x \neq a$ in some neighborhood of the point a , and assume that $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} h(x) = L$.

Then

$$\lim_{x \rightarrow a} g(x) = L.$$

The Squeeze Theorem can sometimes be used to simplify rigorous limit proofs.

Example

Show that $\lim_{x \rightarrow -1} \frac{x^2 - 1}{x + 1} = -2$.

Solution

Let's use the Squeeze Theorem. Letting $x \neq -1$, we obtain

$$0 \leq \left| \frac{x^2 - 1}{x + 1} + 2 \right| = \left| \frac{x^2 + 2x + 1}{x + 1} \right| = \left| \frac{(x + 1)^2}{x + 1} \right| = |x + 1| \xrightarrow{x \rightarrow -1} 0,$$

where the limit $\lim_{x \rightarrow -1} (x + 1) = 0$ follows from the continuity of the polynomial $x \mapsto x + 1$.

Hence $\lim_{x \rightarrow -1} \frac{x^2 - 1}{x + 1} = -2$.

Here is another example illustrating the Squeeze Theorem.

Example

Show that $\lim_{x \rightarrow 0^+} x^2 \sin(\ln(x)) = 0$.

Solution

Let's use the Squeeze Theorem. Letting $x > 0$, we obtain

$$0 \leq |x^2 \sin(\ln(x))| \stackrel{|\sin(\cdot)| \leq 1}{\leq} |x^2| \xrightarrow{x \rightarrow 0^+} 0.$$

Hence $\lim_{x \rightarrow 0^+} x^2 \sin(\ln(x)) = 0$.

Example

The function

$$f(x) = \sin\left(\frac{1}{x}\right)$$

does not have a limit at the origin, as the graph of the function oscillates wildly as it approaches the origin. In contrast, the function

$$g(x) = x \sin\left(\frac{1}{x}\right)$$

has a limit of 0 at the origin, since the factor x tames the oscillation as it approaches the origin. However, the limit cannot be calculated using a suitable simplification or the formulas presented, but it can be done using the Squeeze Theorem.

Since $\mathcal{R}(\sin) = [-1, 1]$, we have $-|x| \leq \sin(\frac{1}{x})x \leq |x|$, so we can take the lower bounding function as $f(x) = -|x|$ and the upper bounding function as $h(x) = |x|$. (You can draw the graphs of all three functions in the same figure for illustration.) Since it also holds that $\lim_{x \rightarrow 0} f(x) = 0 = \lim_{x \rightarrow 0} h(x)$, the assumptions of the previous theorem are satisfied, thus $\lim_{x \rightarrow 0} g(x) = 0$ also holds.

In an earlier Task, it was noted that calculating the instantaneous rate of change of sine can be reduced to determining two trigonometric limits. Let's now go through this.

Example

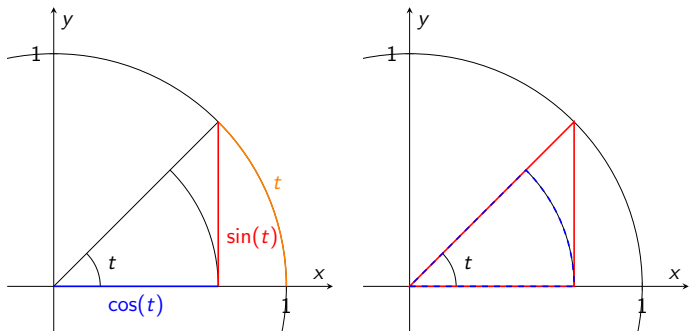
We derive the limit formulas

$$\lim_{t \rightarrow 0} \frac{\sin(t)}{t} = 1 \quad \text{and} \quad \lim_{t \rightarrow 0} \frac{\cos(t) - 1}{t} = 0$$

using the Squeeze Theorem. In the first case, we need to find functions f and h such that $f(t) \leq \frac{\sin(t)}{t} \leq h(t)$ in the neighborhood of the point $t = 0$, and also $\lim_{t \rightarrow 0} f(t) = 1$ and $\lim_{t \rightarrow 0} h(t) = 1$. Finding these is not at all trivial, and the unit circle must be used for assistance.

Example (continued)

From the above first figure



It can be seen, first of all, that $\sin(t) \leq t$ holds for every $t \geq 0$. Thus, it also follows that

$$\frac{\sin(t)}{t} \leq 1.$$

For negative values of the variable t , the corresponding inequality holds (Justify), so surprisingly, the constant function $h(t) = 1$ can serve as the upper bounding function.

Example (continued)

Based on the figure, several area-based estimates can also be made. The second figure illustrates that, particularly for the sector with radius $\cos(t)$ and the right triangle whose sides are $\cos(t)$ and $\sin(t)$, we obtain the estimate in the case $t \in (0, \frac{\pi}{2}]$:

$$\frac{\cos^2(t)t}{2} \leq \frac{\cos(t)\sin(t)}{2},$$

from which it follows the inequality

$$\cos(t) \leq \frac{\sin(t)}{t}.$$

In the case $t \in [-\frac{\pi}{2}, 0)$, a similar inequality can be deduced (Justify), so it holds in the neighborhood of the point $t = 0$. Since $\lim_{t \rightarrow 0} \cos(t) = 1$ also holds, we can choose the lower bounding function as $f(x) = \cos(x)$. Thus, by the Squeeze Theorem, we have $\lim_{t \rightarrow 0} \frac{\sin(t)}{t} = 1$. (Check with the graphs that the lower and upper bounding functions, as well as the estimates used in the intermediate steps, make sense.)

Example (continued)

The proof of the latter limit can be done using the first limit:

$$\begin{aligned}\lim_{t \rightarrow 0} \frac{\cos(t) - 1}{t} &= \lim_{t \rightarrow 0} \frac{(\cos(t) - 1)(\cos(t) + 1)}{t(\cos(t) + 1)} \\&= \lim_{t \rightarrow 0} \frac{\cos^2(t) - 1}{t(\cos(t) + 1)} \\&= \lim_{t \rightarrow 0} \frac{-\sin^2(t)}{t(\cos(t) + 1)} \\&= \lim_{t \rightarrow 0} \frac{\sin(t)}{t} \lim_{t \rightarrow 0} \frac{-\sin(t)}{\cos(t) + 1} \\&= 1 \cdot \frac{\lim_{t \rightarrow 0} (-\sin(t))}{\lim_{t \rightarrow 0} (\cos(t) + 1)} \\&= \frac{0}{2} = 0.\end{aligned}$$

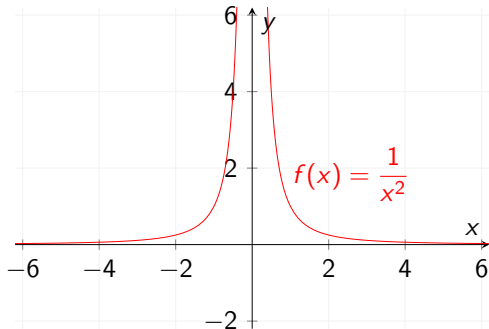
The idea of the first expansion is to get the expression into a form where sine appears.

Exercise

Use the Squeeze Theorem to show that the function $f(x) = -x^2 \cos\left(\frac{1}{x}\right)$ has a limit at the origin.

Infinite Limits ∞ and $-\infty$

The definition of a limit discussed above can be extended to cover cases where the limit is either infinitely large or small (negative), or where the limit is examined for infinitely large or small (negative) values of the variable. For example, the function $f(x) = \frac{1}{x^2}$ has an infinite limit at the point $x = 0$ and a limit of zero at both positive and negative infinity.



In this chapter, we focus on the previously discussed types of limits, while the latter will be addressed in the next chapter.

Definition

Let f be a real-valued function defined about the point $a \in \mathbb{R}$. We say that the function has an **infinite limit** ∞ at the point $x = a$ if for every $M > 0$, there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \implies f(x) > M.$$

(This means that the function $f(x)$ can be made arbitrarily large when the variable x is chosen sufficiently close to the number a from either side.)

In this case we denote $\lim_{x \rightarrow a} f(x) = \infty$ or $f(x) \rightarrow \infty$, as $x \rightarrow a$.

Similarly, we say that the function has an **infinite limit** $-\infty$ at the point $x = a$ if for every $m < 0$, there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \implies f(x) < m.$$

(This means that the value of the function $f(x)$ can be made arbitrarily small (negative) when the variable x is chosen sufficiently close to the number a from either side).

In this case we denote $\lim_{x \rightarrow a} f(x) = -\infty$ or $f(x) \rightarrow -\infty$, as $x \rightarrow a$.

Limits ∞ and $-\infty$ are called **infinite** in contrast to real-valued limits.

One-sided limits ∞ and $-\infty$ are defined similarly to ordinary limits.

Definition

We say that the function f has a left-sided (infinite) limit ∞ at the point $x = a$ if for every $M > 0$, there exists $\delta > 0$ such that

$$a - \delta < x < a \quad \Rightarrow \quad f(x) > M.$$

(This means that the value of the function $f(x)$ can be made arbitrarily large when the variable x is chosen sufficiently close to the number a from the left side.)

In this case, we denote

$$\lim_{x \rightarrow a^-} f(x) = \infty \quad \text{or} \quad f(x) \rightarrow \infty, \text{ as } x \rightarrow a^-.$$

The right-sided (infinite) limit ∞ is defined analogously and denoted by

$$\lim_{x \rightarrow a^+} f(x) = \infty \quad \text{or} \quad f(x) \rightarrow \infty, \text{ as } x \rightarrow a^+.$$

Definition

We say that the function f has a left-sided (infinite) limit $-\infty$ at the point $x = a$ if for every $m < 0$, there exists $\delta > 0$ such that

$$a - \delta < x < a \implies f(x) < m.$$

(This means that the value of the function $f(x)$ can be made arbitrarily small (negative) when the variable x is chosen sufficiently close to the number a from the left side.)

In this case, we denote

$$\lim_{x \rightarrow a^-} f(x) = -\infty \quad \text{or} \quad f(x) \rightarrow -\infty, \text{ as } x \rightarrow a^-.$$

The right-sided (infinite) limit $-\infty$ is defined analogously and denoted by

$$\lim_{x \rightarrow a^+} f(x) = -\infty \quad \text{or} \quad f(x) \rightarrow -\infty, \text{ as } x \rightarrow a^+.$$

As in the case of ordinary limits, limits ∞ and $-\infty$ exist only when the one-sided limits are equal.

Theorem

A function has an (infinite) limit ∞ at the point $x = a$ if and only if its both-sided limits exist and are equal. In other words:

$$\lim_{x \rightarrow a} f(x) = \infty \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = \infty.$$

A function has an (infinite) limit $-\infty$ at the point $x = a$ if and only if its both-sided limits exist and are equal. In other words:

$$\lim_{x \rightarrow a} f(x) = -\infty \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = -\infty.$$

Infinite limits are typically encountered when dividing by zero (or when examining the value of a function as $x \rightarrow \infty$ and $x \rightarrow -\infty$, which will be discussed in the next chapter). Determining the limit in the case of division by zero relies on applying the following result.

Theorem

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty \quad \text{and} \quad \lim_{x \rightarrow 0^+} \frac{1}{x} = \infty.$$

From this, it follows that, for example,

$$\lim_{x \rightarrow 2^+} \frac{1}{\underbrace{x-2}_{\rightarrow 0^+}} = \infty.$$

It is essential in these calculations to consider whether the denominator approaches zero from the negative or positive side, as this affects the sign of infinity. If the denominator approaches some other real number than zero, such one-sided consideration is not needed, since the sign of the denominator will be the same whether approached from the left or the right.

Example

Determine the following infinite limits or show that they do not exist:

$$(a) \lim_{x \rightarrow -2} \frac{x}{x^3 + 8}$$

$$(b) \lim_{x \rightarrow 5} \frac{x^2 + 2x + 1}{(x^2 - 4x - 5)^2} \quad (c) \lim_{x \rightarrow 5} \frac{x^2 + 2x + 1}{x^2 - 4x - 5}.$$

Solution

- (a) Direct substitution $x = -2$ leads to the division $-2/0$, so there may be a limit ∞ or $-\infty$ at this point. This is clarified by calculating the one-sided limits:

$$\lim_{x \rightarrow -2^-} \frac{\overbrace{x}^{\rightarrow -2}}{\underbrace{\underbrace{x^3}_{\rightarrow -8^-} + \underbrace{8}_{\rightarrow 8}}_{\rightarrow 0^-}} = \infty$$

$$\lim_{x \rightarrow -2^+} \frac{\overbrace{x}^{\rightarrow -2}}{\underbrace{\underbrace{x^3}_{\rightarrow -8^+} + \underbrace{8}_{\rightarrow 8}}_{\rightarrow 0^+}} = -\infty$$

The limit does not exist at $x = -2$, since $\lim_{x \rightarrow -2^-} f(x) \neq \lim_{x \rightarrow -2^+} f(x)$.

Solution (continued)

(b)

$$\lim_{x \rightarrow 5^\pm} \frac{\overbrace{x^2 + 2x + 1}^{\rightarrow 36}}{\underbrace{(x^2 - 4x - 5)^2}_{x \rightarrow 0+}} = \infty$$

Answer: $\lim_{x \rightarrow 5} f(x) = \infty$.

Solution (continued)

- (c) Unlike the previous case, with direct substitution we cannot determine from which side the denominator approaches zero, so the expression must be manipulated into a form that reveals the situation. This can be achieved by writing the division in a different form:

$$\lim_{x \rightarrow 5^-} \frac{x^2 + 2x + 1}{\underbrace{x^2 - 4x - 5}_{25^- - 20^+ - 5 = 0^-}} = \lim_{x \rightarrow 5^-} \frac{\overbrace{x^2 + 2x + 1}^{\rightarrow 36}}{\underbrace{(x+1)}_{\rightarrow 6} \underbrace{(x-5)}_{\rightarrow 0^-}} = -\infty,$$

$$\lim_{x \rightarrow 5^+} \frac{x^2 + 2x + 1}{\underbrace{x^2 - 4x - 5}_{25^+ - 20^- - 5 = 0^+}} = \lim_{x \rightarrow 5^+} \frac{\overbrace{x^2 + 2x + 1}^{\rightarrow 36}}{\underbrace{\underbrace{(x+1)}_{\rightarrow 6} \underbrace{(x-5)}_{\rightarrow 0^+}}_{\rightarrow 0^+}} = \infty.$$

The limit does not exist at $x = 5$, since $\lim_{x \rightarrow 5^-} f(x) \neq \lim_{x \rightarrow 5^+} f(x)$.

Basic Exercise

Determine the following limits or show that they do not exist:

$$(a) \lim_{x \rightarrow 1} \frac{1}{x^2 - 1}$$

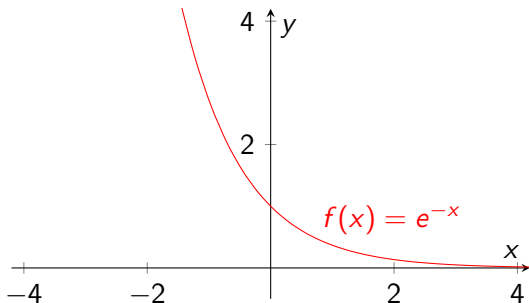
$$(b) \lim_{x \rightarrow 1} \frac{1}{x^2 - 2x + 1}$$

$$(c) \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$$

$$(d) \lim_{x \rightarrow -2} \frac{x + 2}{x^3 + 8}.$$

Limits at Infinity and Negative Infinity

The values of the function $f(x) = e^{-x}$ approach zero as the value of the variable increases to infinity.



This value can also be analyzed using limits. In this case, we are talking about limits at infinity.

Definition

We say that the function has the limit L at infinity ∞ if for every $\varepsilon > 0$, there exists $M > 0$ such that

$$x > M \implies |f(x) - L| < \varepsilon.$$

(This means that the value of a function $f(x)$ can be made as close to the real number L as desired when the variable x is chosen sufficiently large.) In this case, we denote

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad f(x) \rightarrow L, \text{ as } x \rightarrow \infty.$$

Similarly, we say that the function has the limit L at negative infinity $-\infty$ if for every $\varepsilon > 0$, there exists $m < 0$ such that

$$x < m \implies |f(x) - L| < \varepsilon.$$

(This means that the value of a function $f(x)$ can be made as close to the real number L as desired when the variable x is chosen sufficiently small (negative).) In this case, we denote

$$\lim_{x \rightarrow -\infty} f(x) = L \quad \text{or} \quad f(x) \rightarrow L, \text{ as } x \rightarrow -\infty.$$

Often, functions have infinite limits at infinity, such as the function $f(x) = e^{-x}$ at negative infinity $-\infty$.

Definition

We say that the function has a (infinite) limit ∞ at infinity ∞ if for every $M > 0$, there exists $a_M > 0$ such that

$$x > a_M \implies f(x) > M.$$

(This means that the value of a function $f(x)$ can be made arbitrarily large when the variable x is chosen sufficiently large.) In this case, we denote

$$\lim_{x \rightarrow \infty} f(x) = \infty \quad \text{or} \quad f(x) \rightarrow \infty, \text{ as } x \rightarrow \infty.$$

We define a (infinite) limit $-\infty$ at infinity ∞ analogously and denote it as

$$\lim_{x \rightarrow \infty} f(x) = -\infty \quad \text{or} \quad f(x) \rightarrow -\infty, \text{ as } x \rightarrow \infty.$$

Definition

We say that the function has a (infinite) limit ∞ at negative infinity $-\infty$ if for every $M > 0$, there exists $b_M < 0$ such that

$$x < b_M \implies f(x) > M.$$

(This means that the value of a function $f(x)$ can be made arbitrarily large when the variable x is chosen sufficiently small (negative).) In this case, we denote

$$\lim_{x \rightarrow -\infty} f(x) = \infty \quad \text{or} \quad f(x) \rightarrow \infty, \text{ as } x \rightarrow -\infty.$$

The (infinite) limit $-\infty$ at negative infinity $-\infty$ is defined analogously and it is denoted as

$$\lim_{x \rightarrow -\infty} f(x) = -\infty \quad \text{or} \quad f(x) \rightarrow -\infty, \text{ as } x \rightarrow -\infty.$$

A limit or infinite limit at a point $a \in \mathbb{R}$ exists if and only if both one-sided limits are equal, but for the limit at infinity, there is no such condition, as the limit can only be considered from one direction.

The determination of limits at infinity is based on the following limits in the case of rational functions.

Theorem

$$\lim_{x \rightarrow \infty} x^r = \infty \quad \text{and} \quad \lim_{x \rightarrow \infty} \frac{1}{x^r} = 0, \quad r > 0.$$

These limits are easy to accept based on the graphs of power functions (in the case $0 < r < 1$ for root functions). In the case of rational functions, one can also try direct substitution and apply the following equations.

Theorem

$$(1) a \pm \infty = \pm\infty + a = \pm\infty$$

$$(2) a \cdot (\pm\infty) = \pm\infty \cdot a = \begin{cases} \pm\infty, & a > 0 \\ \mp\infty, & a < 0 \end{cases}$$

$$(3) \frac{a}{\pm\infty} = 0$$

$$(4) \frac{\pm\infty}{a} = \begin{cases} \pm\infty, & a > 0 \\ \mp\infty, & a < 0 \end{cases}$$

$$(5) \pm\infty \pm \infty = \pm\infty$$

$$(6) \pm\infty \cdot (\pm\infty) = \infty$$

$$(7) \mp\infty \cdot (\pm\infty) = -\infty.$$

(The signs of the corresponding heights correspond to the same equation, so for example, the notation $\mp\infty \cdot (\pm\infty) = -\infty$ means two equations: $-\infty \cdot \infty = -\infty$ and $\infty \cdot (-\infty) = -\infty$.)

These equations are just a shorthand for limit notations. For example the equation $a + \infty = \infty$ means that if $\lim_{x \rightarrow b} f(x) = a$ and $\lim_{x \rightarrow b} g(x) = \infty$, then $\lim_{x \rightarrow b} (f(x) + g(x)) = \lim_{x \rightarrow b} g(x) = \infty$.

The following expressions involving infinity remain undefined, similar to the division "0/0":

$$(8) \infty - \infty$$

$$(9) 0 \cdot \infty$$

$$(10) \frac{\infty}{\infty}.$$

For example, expression (8) is undefined because the limits $\lim_{x \rightarrow \infty} (x^2 - x)$, $\lim_{x \rightarrow \infty} (x - x)$, and $\lim_{x \rightarrow \infty} (x - x^2)$ all lead to this form, even though their values are different (You can deduce this based on the graphs). Thus, the difference $\infty - \infty$ cannot be assigned a value that would work consistently.

In these cases, the expression of a rational function must be manipulated into a form that does not yield undefined expressions upon substitution, as in the earlier case of the expression “0/0”. Generally, this can be accomplished by dividing both the numerator and denominator by the highest power of the variable x^n , which will be familiar from the next example. The resulting expression is not equal to the original at the point $x = 0$, but this is not significant because the limit is not considered at that point.

Example

Determine the following limits:

$$(a) \lim_{x \rightarrow -\infty} \frac{2}{3x - 4}$$

$$(c) \lim_{x \rightarrow \pm\infty} \frac{2x^2 - x + 3}{3x^2 + 5}$$

$$(b) \lim_{x \rightarrow \pm\infty} \frac{3x - 4}{2}$$

$$(d) \lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x).$$

Solution

(a)

$$\lim_{x \rightarrow -\infty} \frac{2}{3x - 4} = \lim_{x \rightarrow -\infty} \frac{\frac{2}{x}}{3 - \frac{4}{x}} = \frac{0}{3 - 0} = 0$$

or alternatively by direct substitution

$$\lim_{x \rightarrow -\infty} \frac{2}{3x - 4} = \frac{2}{3 \cdot (-\infty) - 4} = \frac{2}{-\infty - 4} = \frac{2}{-\infty} = 0.$$

(b)

$$\lim_{x \rightarrow \pm\infty} \frac{3x - 4}{2} = \lim_{x \rightarrow \pm\infty} \frac{3 - \overbrace{\frac{4}{x}}^{\rightarrow 0}}{\underbrace{2}_{\rightarrow 0^\pm}} = \pm\infty$$

or alternatively by direct substitution

$$\lim_{x \rightarrow \pm\infty} \frac{3x - 4}{2} = \frac{\pm\infty - 4}{2} = \frac{\pm\infty}{2} = \pm\infty.$$

Solution (continued)

(c)

$$\lim_{x \rightarrow \pm\infty} \frac{2x^2 - x + 3}{3x^2 + 5} = \lim_{x \rightarrow \pm\infty} \frac{2 - \frac{1}{x} + \frac{3}{x^2}}{3 + \frac{5}{x^2}} = \frac{2 - 0 + 0}{3 + 0} = \frac{2}{3}$$

(d)

$$\begin{aligned} & \lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x) && | = \frac{0}{0} \\ &= \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + x} - x)(\sqrt{x^2 + x} + x)}{\sqrt{x^2 + x} + x} \\ &= \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + x})^2 - x^2}{\sqrt{x^2 + x} + x} \\ &= \lim_{x \rightarrow \infty} \frac{x^2 + x - x^2}{\sqrt{x^2 + x} + x} \\ &= \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + x} + x} && | = \frac{0}{0} \\ &= \lim_{x \rightarrow \infty} \frac{1}{\frac{1}{x}\sqrt{x^2 + x} + 1} = \dots \end{aligned}$$

Solution (continued)

$$\begin{aligned}\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x) &= \lim_{x \rightarrow \infty} \frac{1}{\frac{1}{x}\sqrt{x^2 + x} + 1} && | = "0 \cdot \infty" \\&= \lim_{x \rightarrow \infty} \frac{1}{\frac{1}{x}\sqrt{x^2 \left(1 + \frac{1}{x}\right)} + 1} \\&= \lim_{x \rightarrow \infty} \frac{1}{\frac{|x|}{x}\sqrt{1 + \frac{1}{x}} + 1} && | x > 0 \\&= \lim_{x \rightarrow \infty} \frac{1}{\frac{x}{x}\sqrt{1 + \frac{1}{x}} + 1} \\&= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{x}} + 1} \\&= \frac{1}{\sqrt{1 + 0} + 1} = \frac{1}{2}.\end{aligned}$$

Limits at infinity can also be calculated using an alternative arrow notation for limits, which allows us to omit the limit notation. For example, part (c) can be expressed as

$$\frac{2x^2 - x + 3}{3x^2 + 5} = \frac{2 - \frac{1}{x} + \frac{3}{x^2}}{3 + \frac{3}{x^2}} \xrightarrow{x \rightarrow \pm\infty} \frac{2 - 0 + 0}{3 + 0} = \frac{2}{3}$$

or in the form

$$\frac{2x^2 - x + 3}{3x^2 + 5} = \frac{2 - \frac{1}{x} + \frac{3}{x^2}}{3 + \frac{3}{x^2}} \rightarrow \frac{2 - 0 + 0}{3 + 0} = \frac{2}{3}, \quad \text{as } x \rightarrow \pm\infty.$$

However, these notations should not be mixed. For example,

$\lim_{x \rightarrow \infty} f(x) \rightarrow L$ is an incorrect notation.

Note that in the case of rational functions, based on the highest powers of the numerator and denominator, one can deduce whether the limit at infinity results in 0 , $\pm\infty$, or some non-zero real number. Try to infer this in the following exercise before calculating the limits.

Basic Exercise

Determine the following limits:

$$(a) \lim_{x \rightarrow \pm\infty} \frac{3x^2 + 2x + 1}{4 + 3x + 2x^2 + x^3}$$

$$(b) \lim_{x \rightarrow \pm\infty} \frac{3 + 2x + x^2}{3x^2 + 2x + 1}$$

$$(c) \lim_{x \rightarrow \pm\infty} \frac{4 + 3x + 2x^2 + x^3}{3x^2 + 2x + 1}.$$

Lastly, let's specifically mention the limit of the exponential function at infinity.

Theorem

In the case where $0 < a < 1$, it holds that

$$\lim_{x \rightarrow \infty} a^x = 0 \quad \text{and} \quad \lim_{x \rightarrow -\infty} a^x = \infty$$

and in the case where $a > 1$,

$$\lim_{x \rightarrow \infty} a^x = \infty \quad \text{and} \quad \lim_{x \rightarrow -\infty} a^x = 0.$$

In cases where $a < 0$ and $a = 1$, the exponential function is not defined.

The previous limits should be quite clear for the case $x \rightarrow \infty$, and for the case $x \rightarrow -\infty$, it is helpful to consider them through the reciprocal $a^{-x} = 1/a^x$. The corresponding limits for other basic functions are left as an exercise.

Basic Exercise

Verify (for example, using a graph) that you know the following limits or that they do not exist:

$$(a) \lim_{x \rightarrow \pm\infty} x^n, n \in \{1, 3, 5, \dots\}$$

$$(b) \lim_{x \rightarrow \pm\infty} -x^n, n \in \{1, 3, 5, \dots\}$$

$$(c) \lim_{x \rightarrow \pm\infty} x^n, n \in \{2, 4, 6, \dots\}$$

$$(d) \lim_{x \rightarrow \pm\infty} -x^n, n \in \{2, 4, 6, \dots\}$$

$$(e) \lim_{x \rightarrow \pm\infty} \sqrt[n]{x}, n \in \{3, 5, 7, \dots\}$$

$$(f) \lim_{x \rightarrow \infty} \sqrt[n]{x}, n \in \{2, 4, 6, \dots\}$$

$$(g) \lim_{x \rightarrow \pm\infty} \frac{1}{x}$$

$$(h) \lim_{x \rightarrow \pm\infty} \sin(x)$$

$$(i) \lim_{x \rightarrow \pm\infty} \cos(x)$$

$$(j) \lim_{x \rightarrow \pm\infty} \tan(x)$$

$$(k) \lim_{x \rightarrow \pm\infty} e^x$$

$$(l) \lim_{x \rightarrow \pm\infty} e^{-x}$$

$$(m) \lim_{x \rightarrow \infty} \ln(x)$$

$$(n) \lim_{x \rightarrow 0^+} \ln(x)$$

$$(o) \lim_{x \rightarrow \infty} \log_{\frac{1}{2}}(x) \quad (p) \lim_{x \rightarrow 0^+} \log_{\frac{1}{2}}(x).$$

L'Hôpital's Rule

L'Hôpital's rule is a method based on differentiation that can be used to calculate the limit of a function in certain cases.

Example

Determine the limit

$$\lim_{x \rightarrow 0} \frac{x}{\sin(x)}.$$

Solution

Calculating the limit directly does not work, as this leads to the indeterminate expression “0/0”. In this case, we can try differentiating both the numerator and the denominator and then compute the limit:

$$\lim_{x \rightarrow 0} \frac{x}{\sin(x)} = \lim_{x \rightarrow 0} \frac{1}{\cos(x)} = \frac{1}{\cos(0)} = 1.$$

Success! Thus, $\lim_{x \rightarrow 0} \frac{x}{\sin(x)} = 1$.

The first equality in the solution of the previous example is based on L'Hôpital's rule, which can be used not only in the case of "0/0" but also in the case of " ∞/∞ ".

Theorem (L'Hôpital's Rule)

Let

- (1) functions f and g be differentiable on some interval containing the number c , specifically (a, b) , $(-\infty, b)$, or (a, ∞) , possibly excluding the point c .
- (2) $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = 0$ or $\lim_{x \rightarrow c} |f(x)| = \lim_{x \rightarrow c} |g(x)| = \infty$.
- (3) $g'(x) \neq 0$ on the interval (a, b) , $(-\infty, b)$, or (a, ∞) , possibly excluding the point c .
- (4) the limit $\lim_{x \rightarrow c} (f'(x)/g'(x))$ exists, either finite or infinite.

Then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}.$$

The theorem also applies to one-sided limits $x \rightarrow c^-$ and $x \rightarrow c^+$ when c is a finite number.

The various assumptions of the theorem may make the rule seem daunting, but its application is actually quite practical, as demonstrated in the first example. The satisfaction of assumption (2) is typically noted in practice when the limit cannot be computed by direct substitution, and the fulfillment of assumption (4) is seen when attempting to calculate the limit after differentiation.

Assumption (1) usually holds true for the typical functions encountered (for example, choose $(a, b) = (-0.1, 0.1)$ in the previous example).

Assumption (3), on the other hand, means that if $c \in \mathbb{R}$, then the derivative g' must not have a zero arbitrarily close to the point c , and in the cases where $c = \pm\infty$, that the derivative g' must not have arbitrarily large/small zeros.

However, assumption (2) is critical, so let's illustrate its necessity. Clearly,

$$\lim_{x \rightarrow 0} \frac{x}{1+x} = 0,$$

but using L'Hôpital's rule would yield

$$\lim_{x \rightarrow 0} \frac{x}{1+x} = \lim_{x \rightarrow 0} \frac{1}{1} = 1.$$

This contradiction arises because assumption (2) does not hold in this case.

Let's take another example where L'Hôpital's rule can also be used to calculate limits at infinity. It's also important to note that the rule can be applied multiple times in succession. In this case, it is essential to ensure that assumption (2) holds each time.

Example

Determine the limit

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^2}.$$

Solution

$$\lim_{x \rightarrow \infty} \frac{\overbrace{e^x}^{\frac{\infty}{\infty}}}{x^2} = \lim_{x \rightarrow \infty} \frac{\overbrace{e^x}^{\frac{\infty}{\infty}}}{2x} = \lim_{x \rightarrow \infty} \frac{e^x}{2} = \infty.$$

The result of the previous example can be repeated in any case of the form e^x/x^n , where $n \in \mathbb{Z}_{\geq 0}$. In fact, it can be shown that this also holds for the case $n \in \mathbb{R}$.

Theorem

The exponential function e^x grows faster than any polynomial function.

Basic Exercise

Determine the following limits:

$$(a) \lim_{x \rightarrow 0} \frac{\sin(x)}{x}$$

$$(b) \lim_{x \rightarrow \infty} \frac{x^3}{e^x}.$$

The limit for part (a) is needed to define the derivative of sine. However, it cannot be determined for this purpose using L'Hôpital's rule since the derivative must already be known when using the rule. Therefore, this limit was determined in Chapter 1.1.8 using the squeeze theorem, allowing the derivative of sine to be deduced without falling into circular reasoning.

The result of part (b) of the previous exercise can also be repeated in any case of the form x^n/e^x , where $n \in \mathbb{Z}_{\geq 0}$, and in fact, it also holds for the case $n \in \mathbb{R}$, which means that “the function e^{-x} approaches zero faster than any polynomial function as x approaches infinity”.

Continuity

A function f is continuous on the interval $[a, b]$ if and only if the following holds: $\lim_{x \rightarrow c} f(x) = f(c)$ for every $c \in (a, b)$, and $\lim_{x \rightarrow a^+} f(x) = f(a)$ and $\lim_{x \rightarrow b^-} f(x) = f(b)$.

From the definition of continuity, we can conclude that: Let f be continuous on the closed interval $[a, b]$. Then

- 1) **Maximum-Minimum Theorem:** f attains
 - a maximum and a minimum value,
 - all values in between
- 2) **Intermediate Value Theorem:** if s is between $f(a)$ and $f(b)$, then there exists $c \in [a, b]$ such that $f(c) = s$

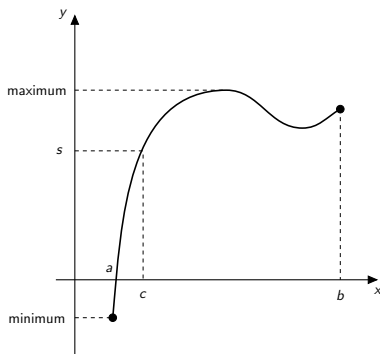


Figure: A figure illustrating the Intermediate Value Theorem.

It is important to note that the above discussion dealt with continuity on a closed interval. If it is only known that the function f is continuous on the interval (a, b) , then f may behave quite peculiarly near the points a and b .

Extending a Function to be Continuous. If the limit $\lim_{x \rightarrow a} f(x)$ exists as a finite number but $a \notin \mathcal{D}(f)$, meaning f is not defined at the point a , a natural idea is to make an extension

$$F(x) = \begin{cases} f(x), & x \in \mathcal{D}(f) \\ \lim_{t \rightarrow a} f(t), & x = a \end{cases}.$$

Then, if $f(x)$ was continuous on its domain, $F(x)$ will also be continuous.

Example

If possible, extend the following functions to be continuous over $\mathbb{R}$:

$$(a) f(x) = \frac{x^2 - x}{x - 1}$$

$$(b) g(x) = \begin{cases} \sqrt[3]{x}, & x < 4 \\ \frac{x}{10} + \frac{6}{5}, & x > 4 \end{cases}.$$

Solution

We need to identify the points where the function is not defined and assign to each such point $x = a$ a value such that

$$\lim_{x \rightarrow a} f(x) = f(a)$$

that is,

$$\lim_{x \rightarrow a^-} f(x) = f(a) = \lim_{x \rightarrow a^+} f(x).$$

- (a) The domain of the function f is $\mathcal{D}(f) = \mathbb{R} \setminus \{1\}$. The limit at the point $x = 1$ is

$$\lim_{x \rightarrow 1} \frac{x^2 - x}{x - 1} = \lim_{x \rightarrow 1} \frac{x(x - 1)}{(x - 1) \cdot 1} = \lim_{x \rightarrow 1} x = 1.$$

For the extended function h , it must hold that $h(1) = 1$. Thus, the function

$$h(x) = \begin{cases} f(x), & x \neq 1 \\ \lim_{t \rightarrow 1} f(t), & x = 1 \end{cases} = \begin{cases} \frac{x^2 - x}{x - 1}, & x \neq 1 \\ 1, & x = 1 \end{cases}$$

is continuous over $\mathbb{R}$.

Solution (continued)

- (b) The function g is not defined at the point $x = 4$. Since its expression changes at this point, we must check if the one-sided limits are equal.

$$\lim_{x \rightarrow 4^-} g(x) = \lim_{x \rightarrow 4} \sqrt[3]{x} = \sqrt[3]{4} \approx 1.587$$

but

$$\lim_{x \rightarrow 4^+} g(x) = \lim_{x \rightarrow 4} \frac{x}{10} + \frac{6}{5} = \frac{8}{5} = 1.6,$$

so the function does not have a limit at this point, and it cannot be extended to be continuous.

Basic Exercise

If possible, extend the following functions to be continuous at their points of discontinuity:

$$(a) f(x) = \frac{4 - x^2}{3x - 6}$$

$$(c) f(x) = \frac{1}{x}$$

$$(b) f(x) = \begin{cases} \frac{x^2 - 8x - 1}{x - 3}, & x < 2 \\ 2^{x+2} - 3, & x > 2 \end{cases}$$

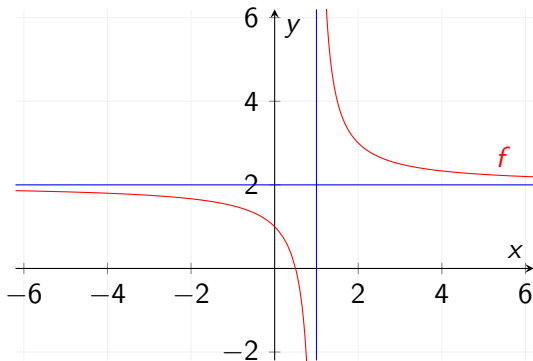
$$(d) f(x) = \frac{|x - 2|}{x^2 + x - 6}.$$

Provide the extended functions.

Exercise

Extend $f(x) = \frac{\sin(x)}{x}$ to be continuous over $\mathbb{R}$.

Let's begin by examining the accompanying figure.



The graph of the function in the figure resembles the line $y = 2$ for arbitrarily large and small (negative) values of the variable x , and the line $x = 1$ when x is arbitrarily close to the value 1. Such a line is called an asymptote.

Definition

A function f has a **vertical asymptote** at the point $x = a$ if at least one of the following conditions holds:

$$\lim_{x \rightarrow a^-} f(x) = -\infty, \quad \lim_{x \rightarrow a^-} f(x) = \infty, \quad \lim_{x \rightarrow a^+} f(x) = -\infty \quad \text{or} \quad \lim_{x \rightarrow a^+} f(x) = \infty$$

a **horizontal asymptote** $y = L$ if

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L$$

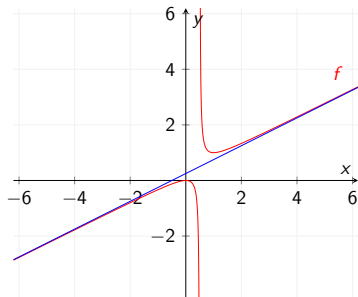
or both, and an **oblique asymptote** $y = ax + b$ if

$$\lim_{x \rightarrow \infty} (f(x) - (ax + b)) = 0 \quad \text{or} \quad \lim_{x \rightarrow -\infty} (f(x) - (ax + b)) = 0$$

or both.

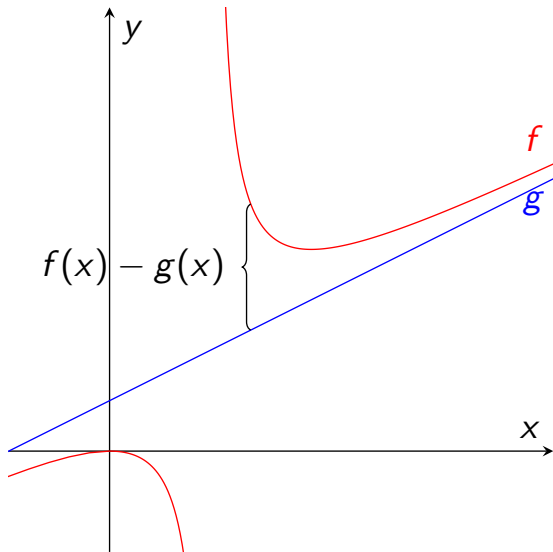
The definitions of vertical and horizontal asymptotes are quite evident: a vertical asymptote exists at a point where the function's value approaches infinity, and a horizontal asymptote exists if the function has a (finite) limit. Let's examine the definition of oblique asymptotes a bit more closely.

In the accompanying figure, the graph of the function $f(x) = \frac{x^2}{2x-1}$ and its asymptote $y = \frac{x}{2} + \frac{1}{4}$ are shown.



The function f behaves like the function $g(x) = \frac{x}{2} + \frac{1}{4}$ for large values of the variable. One might quickly and incorrectly think that $\lim_{x \rightarrow \infty} f(x) = g(x)$. However, this is not the case, since the limit in question becomes infinite, not $\frac{x}{2} + \frac{1}{4}$.

Instead, the relationship between the functions f and g can be expressed by stating that the **difference** $f(x) - g(x)$ approaches zero as the variable x increases arbitrarily large. This is precisely the definition of an oblique asymptote. This concept is further illustrated in the next figure.



Vertical asymptotes can be as many as needed. In contrast, oblique and horizontal asymptotes can be at most two: one that describes the behavior of the function for small (negative) values of the variable, and another that describes the behavior of the function for large values of the variable. This is because the value of the function is uniquely defined. For example, the function $\arctan(x)$ has two horizontal asymptotes, and the function

$$f(x) = \begin{cases} e^x, & x < 1 \\ \frac{x^2}{2x-1}, & x \geq 1 \end{cases}$$

has both a horizontal and an oblique asymptote. However, in this course, we will restrict ourselves to calculating only the asymptotes of rational functions, which can have at most one horizontal or oblique asymptote. Furthermore, the vertical asymptotes always occur at points where division by zero would take place. However, dividing by zero does not directly imply that there is a vertical asymptote at that point (give a counterexample), but this must be verified through calculation.

In practical applications, horizontal asymptotes occur quite often. Often, a system approaches some equilibrium state over time. This is the case, for example, with the concentrations of solutions in mixing tanks (e.g., salt solution). On the other hand, if the quantity under consideration is transformed to another scale, it may have an oblique asymptote, the slope of which may be of interest.

Example

Find the asymptotes for the following functions:

$$(a) \ f(x) = \frac{(x+1)(x-2)}{x^2-1}$$

$$(b) \ g(x) = \frac{x^3 - 2x - 3}{x^2 + 1}.$$

Solution

(a) Vertical asymptotes:

$x = 1$:

$$\lim_{x \rightarrow 1^+} \frac{\overbrace{(x+1)(x-2)}^{\rightarrow -2}}{\underbrace{x^2 - 1}_{\rightarrow 0^+}} = -\infty$$

thus, $x = 1$ is an asymptote.

$x = -1$:

$$\lim_{x \rightarrow -1} \frac{(x+1)(x-2)}{x^2 - 1} = \lim_{x \rightarrow -1} \frac{(x+1)(x-2)}{(x+1)(x-1)} = \lim_{x \rightarrow -1} \frac{x-2}{x-1} = \frac{-1-2}{-1-1} = \frac{3}{2},$$

no asymptote.

Horizontal asymptote:

$$\lim_{x \rightarrow \pm\infty} \frac{(x+1)(x-2)}{x^2 - 1} = \dots = \lim_{x \rightarrow \pm\infty} \frac{x-2}{x-1} = \lim_{x \rightarrow \pm\infty} \frac{1 - \frac{2}{x}}{1 - \frac{1}{x}} = \frac{1}{1} = 1,$$

thus, $y = 1$ is an asymptote (and hence there cannot be an oblique asymptote).

Answer: $x = 1$ and $y = 1$.

Solution (continued)

(b) The function g cannot have vertical asymptotes (Why not?).

Oblique asymptote:

Method 1: It should hold that

$$\begin{aligned} 0 &= \lim_{x \rightarrow \pm\infty} \left(\frac{x^3 - 2x - 3}{x^2 + 1} - (ax + b) \right) \\ &= \lim_{x \rightarrow \pm\infty} \left(\frac{x^3 - 2x - 3 - (ax + b)(x^2 + 1)}{x^2 + 1} \right) \\ &= \lim_{x \rightarrow \pm\infty} \left(\frac{(1-a)x^3 - bx^2 - (2+a)x - 3-b}{x^2 + 1} \right) \\ &= \lim_{x \rightarrow \pm\infty} \left(\frac{(1-a)x - b - \frac{2+a}{x} - \frac{3+b}{x^2}}{1 + \frac{1}{x^2}} \right), \end{aligned}$$

thus

$$\begin{cases} 1-a=0 \\ -b=0 \end{cases} \iff \begin{cases} a=1 \\ b=0 \end{cases}.$$

Answer: $y = x$.

Solution (continued)

(b) **Method 2:** By using polynomial long division or by performing the division in the form

$$\frac{x^3 - 2x - 3}{x^2 + 1} = \frac{(x^2 + 1)x - x - 2x - 3}{x^2 + 1} = x - 3\frac{x + 1}{x^2 + 1},$$

(where zero was added to the numerator in the form $x - x$ to obtain the term $(x^2 + 1)x$), it is seen that the function g appears to resemble the line $y = x$ for large values of the variable x . (Why can't we just calculate the limit $\lim_{x \rightarrow \pm\infty}$ in the previous examination?). Let's also show using the definition that this is indeed the case:

$$\begin{aligned}\lim_{x \rightarrow \pm\infty} \left(\frac{x^3 - 2x - 3}{x^2 + 1} - x \right) &= \lim_{x \rightarrow \pm\infty} \left(\frac{x^3 - 2x - 3 - x(x^2 + 1)}{x^2 + 1} \right) \\ &= \lim_{x \rightarrow \pm\infty} \left(\frac{-3x - 3}{x^2 + 1} \right) = \lim_{x \rightarrow \pm\infty} \left(\frac{-\frac{3}{x} - \frac{3}{x^2}}{1 + \frac{1}{x^2}} \right) = \frac{0}{1} = 0,\end{aligned}$$

thus the function g has an oblique asymptote $y = x$.

In method 2 of part (b) in the previous example, the asymptote was obtained through division. More generally, it holds that in the case of rational functions, an oblique/horizontal asymptote can be determined by finding the quotient in the polynomial division. However, there still needs to be a justification for why this is sufficient.

A vertical asymptote could also be determined without limits as follows: If the expression for the function at a point x takes the form $\frac{a}{0}$, where $a \neq 0$, then the function has a vertical asymptote at that point. In the case $\frac{0}{0}$, the expression for the function must be simplified further until it takes either the form b or $\frac{a}{0}$, where the first case indicates a hole in the graph and the second case a vertical asymptote.

Note that for vertical asymptotes, it is sufficient to examine only one one-sided limit if it results in $\pm\infty$. If it does not, both one-sided limits must be considered. If the goal is to sketch the graph of the function using the asymptotes, it is necessary to determine both limits to understand how the function behaves on either side of the discontinuity. Also, note that examining a slant asymptote simultaneously clarifies the existence of a horizontal asymptote, since a horizontal asymptote is also a line.

Basic Exercise

Determine the horizontal asymptote of the function from part (a) of the previous example using the method from part (b) 1.

Basic Exercise

Determine the asymptotes of the following functions and sketch the graphs utilizing them:

(a)

$$(i) f(x) = \frac{1}{x-1}$$

$$(ii) g(x) = \frac{x^3 + 6x^2 + 11x + 6}{x^2 - 1}$$

(b) Determine the asymptotes of the following functions (no justification needed by calculation):

$$(i) e^x$$

$$(ii) e^{-x}$$

$$(iii) \ln(x)$$

$$(iv) \arctan(x)$$

$$(v) 4x + e^x.$$

The graphs of functions can behave like curves other than lines for large and small values of the variable. These are called **curvilinear** or **nonlinear asymptotes** as opposed to ordinary linear asymptotes. They can be defined similarly to slant asymptotes: a function f has a curvilinear asymptote g if it holds that

$$\lim_{x \rightarrow \infty} (f(x) - g(x)) = 0 \quad \text{or} \quad \lim_{x \rightarrow -\infty} (f(x) - g(x)) = 0,$$

(where g is not a linear function). For instance, if the asymptote is a quadratic function, one must write $g(x) = ax^2 + bx + c$ in the previous equation.

To determine the curvilinear asymptote with the previous formula, it is necessary to know the type of function the asymptote represents, in order to write the expression for the function g in the correct form. In the case of a rational function $f(x) = p(x)/q(x)$, the asymptote is a polynomial whose degree is the difference of the degrees of the polynomials p and q . If the difference is negative, the asymptote is $y = 0$. For example, if the degree of the polynomial p is 5 and the degree of the polynomial q is 2, then the asymptote of the function $f(x) = p(x)/q(x)$ is a 3rd degree polynomial, and the asymptote of the function $g(x) = q(x)/p(x)$ is $y = 0$.

For instance, if the degree of the denominator is n and the degree of the numerator is $n + 1$, we can write

$$\begin{aligned}
 f(x) &= \frac{a_{n+1}x^{n+1} + a_nx^n + \cdots + a_1x + a_0}{b_nx^n + b_{n-1}x^{n-1} + \cdots + b_1x + b_0} \\
 &= \frac{\frac{a_{n+1}}{b_n}x(b_nx^n + b_{n-1}x^{n-1} + \cdots + b_1x + b_0) + (c_nx^n + c_{n-1}x^{n-1} + \cdots + c_1x + c_0)}{b_nx^n + b_{n-1}x^{n-1} + \cdots + b_1x + b_0} \\
 &= \frac{a_{n+1}}{b_n}x + \frac{c_nx^n + c_{n-1}x^{n-1} + \cdots + c_1x + c_0}{b_nx^n + b_{n-1}x^{n-1} + \cdots + b_1x + b_0} \\
 &= \frac{a_{n+1}}{b_n}x + \frac{c_n + \frac{c_{n-1}}{x} + \cdots + \frac{c_1}{x^{n-1}} + \frac{c_0}{x^n}}{b_n + \frac{b_{n-1}}{x} + \cdots + \frac{b_1}{x^{n-1}} + \frac{b_0}{x^n}},
 \end{aligned}$$

from which we observe that for large values of the variable x , f resembles a first-degree polynomial (On what basis?).

Example

Determine the curvilinear asymptotes of the function

$$f(x) = \frac{x^4}{x+1}$$

Solution

The curvilinear asymptote is clearly a third-degree polynomial based on the degrees of the numerator and denominator. It must then hold that

$$\begin{aligned} 0 &= \lim_{x \rightarrow \pm\infty} \left(\frac{x^4}{x+1} - (ax^3 + bx^2 + cx + d) \right) \\ &= \lim_{x \rightarrow \pm\infty} \frac{x^4 - ax^4 - ax^3 - bx^3 - bx^2 - cx^2 - cx - dx - d}{x+1} \\ &= \lim_{x \rightarrow \pm\infty} \frac{(1-a)x^4 - (a+b)x^3 - (b+c)x^2 - (c+d)x - d}{x+1}. \end{aligned}$$

Solution (continued)

Hence

$$\begin{cases} 1 - a = 0 \\ a + b = 0 \\ b + c = 0 \\ c + d = 0 \end{cases} \iff \begin{cases} a = 1 \\ b = -1 \\ c = 1 \\ d = -1 \end{cases}.$$

Thus, the curvilinear asymptote is $g(x) = x^3 - x^2 + x - 1$.

Let's verify the answer by determining the quotient of the function f :

$$\begin{aligned} f(x) &= \frac{x^4}{x+1} = \frac{(x+1)x^3 - x^3}{x+1} = x^3 + \frac{-x^3}{x+1} \\ &= x^3 + \frac{(x+1)(-x^2) + x^2}{x+1} = x^3 - x^2 + \frac{(x+1)x - x}{x+1} \\ &= x^3 - x^2 + x + \frac{(x+1)(-1) + 1}{x+1} \\ &= x^3 - x^2 + x - 1 + \frac{1}{x+1}. \end{aligned}$$

Basic Exercise

Consider a rational function $f(x) = p(x)/q(x)$, where the degree of the numerator p is m and the degree of the denominator q is n .

- (a) What type of non-vertical asymptote (i.e., horizontal, slant, or curvilinear) does the function f have when $m = 2$ and $n = 2$?
- (b) What type of non-vertical asymptote does the function f have when $m = 2$ and $n = 3$?
- (c) What type of non-vertical asymptote does the function f have when $m < n$?
- (d) What type of non-vertical asymptote does the function f have when $m = 4$ and $n = 2$?
- (e) What type of non-vertical asymptote does the function f have when $m - n = 4$?

Exercise

Determine the asymptotes and curvilinear asymptotes of the function $f(x) = \frac{x^3+1}{2x-1}$.

Derivative

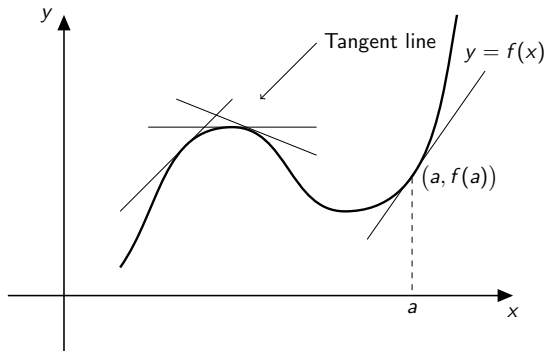


Figure: Tangent lines.

The rate of change of a continuous function's values, or the steepness of the function's graph, can be measured by the slope of the tangent (if it exists): the larger the slope of the tangent, the steeper the graph of the function rises (see above Figure).

The slope of the tangent can be obtained through the secant, which is the line drawn through two points on the function. The slope of the secant over the interval $[x, x + h]$ is

$$\frac{\Delta y}{\Delta x} = \frac{f(x + h) - f(x)}{(x + h) - x} = \frac{f(x + h) - f(x)}{h},$$

and this expression is also called the **difference quotient**. It expresses the **average rate of change** of the function over the interval $[x, x + h]$. As h decreases, the slope of the secant approaches the slope of the tangent, which is the **instantaneous rate of change** of the function. The exact value of the slope of the tangent is obtained by taking the limit $\lim_{h \rightarrow 0}$ of the difference quotient. The function thus obtained

$$\boxed{f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}}, \quad (3)$$

is called the **derivative** of the function f , and it indicates the rate of change of the function f at each value of the variable x . The derivative exists at those points where the previous limit exists. In this case, we say that f is **differentiable** at the point x . If f is differentiable at every point in the interval (a, b) , we say that f is differentiable over the interval (a, b) .

Example

The derivative of the function $f(x) = x^2$ at the point $x = 1$ is defined as

$$\begin{aligned} f'(1) &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(1+h)^2 - 1^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{1^2 + 2h + h^2 - 1}{h} \\ &= \lim_{h \rightarrow 0} \frac{2h + h^2}{h} \\ &= \lim_{h \rightarrow 0} (2 + h) \\ &= 2 + 0 \\ &= 2. \end{aligned}$$

Example (continued)

The derivative of the function at an arbitrary point x is defined as

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\&= \lim_{h \rightarrow 0} \frac{2hx + h^2}{h} \\&= \lim_{h \rightarrow 0} (2x + h) \\&= 2x.\end{aligned}$$

This reasoning works for every $x \in \mathbb{R}$, so the obtained formula $f'(x) = 2x$ holds for every $x \in \mathbb{R}$. For example, based on the formula, we have $f'(1) = 2 \cdot 1 = 2$, which is consistent with the previous calculation.

In practice, derivatives are not computed based on the definition, but rather by using formulas derived from the definition. This will be practiced in a later chapter. In some cases, however, it is necessary to calculate the derivative using the definition, which is addressed in the following exercise.

Example

Determine the derivative of the function $f(x) = |x|$.

Solution

We want to determine the derivative of the function

$$f(x) = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

at each point where it is defined. The situation becomes slightly more complicated than in the previous example due to the fact that the function is piecewise defined.

Solution (continued)

When $x > 0$, we can calculate as in the previous example:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h) - x}{h} \\ &= \lim_{h \rightarrow 0} \frac{h}{h} \\ &= \lim_{h \rightarrow 0} 1 \\ &= 1. \end{aligned}$$

Solution (continued)

In the case $x < 0$, we can calculate similarly:

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{-(x+h) - (-x)}{h} \\&= \lim_{h \rightarrow 0} \frac{-h}{h} \\&= \lim_{h \rightarrow 0} -1 \\&= -1.\end{aligned}$$

However, in the case $x = 0$, the situation is fundamentally different. This is because the expression $f(0+h)$ means the lower branch of the function for $h < 0$, i.e., the expression $-(0+h)$, and the upper branch of the function for $h > 0$, i.e., the expression $0+h$. Therefore, the case $x = 0$ cannot be computed with a single limit, but we must examine the one-sided limits separately. (If you try to calculate with a single limit, you will notice that you do not know what to substitute for the expression $f(0+h)$.)

Solution (continued)

In the case $h > 0$, we have

$$\begin{aligned}f'(0^-) &= \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{-(0+h) - 0}{h} \\&= \lim_{h \rightarrow 0^-} \frac{-h}{h} = \lim_{h \rightarrow 0^-} -1 = -1.\end{aligned}$$

And in the case $h > 0$,

$$\begin{aligned}f'(0^+) &= \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{(0+h) - 0}{h} \\&= \lim_{h \rightarrow 0^+} \frac{h}{h} = \lim_{h \rightarrow 0^+} 1 = 1.\end{aligned}$$

Since the one-sided limits are different, the limit $\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$ or the derivative $f'(0)$ does not exist. Thus, the derivative of the function f is

$$f'(x) = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases},$$

meaning its domain is smaller than that of the original function.

Exercise

Determine the derivative of the function

$$f(x) = \begin{cases} 0, & x < 0 \\ x^2, & x \geq 0 \end{cases}$$

From the definition of the derivative, it follows that the equation of the tangent line to the function f at the point $x = a$ is

$$y - f(a) = f'(a)(x - a)$$

and the equation of the normal line is

$$y - f(a) = -\frac{1}{f'(a)}(x - a)$$

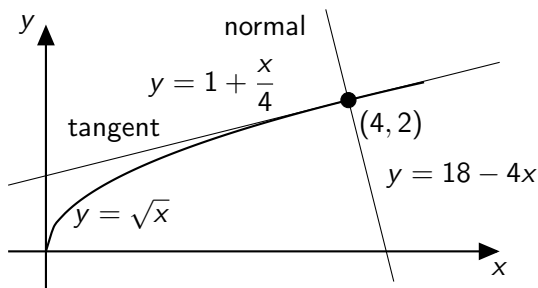


Figure: Tangent and normal.

In the Figure below, typical cases of functions that are not differentiable at a certain point (in this case, at the origin) are presented.

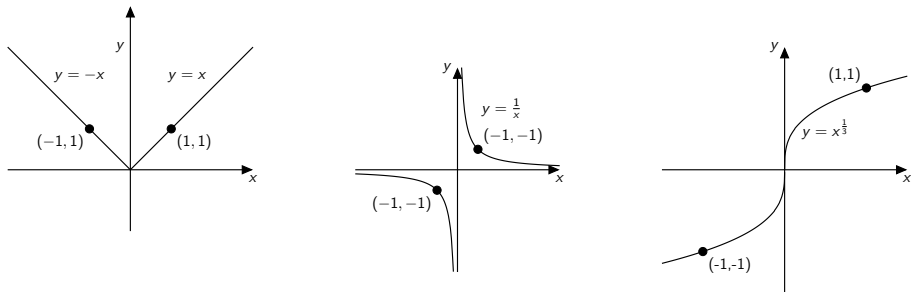


Figure: Functions which are not differentiable at the origin.

(1) No unique tangent (corner):

$$f'_-(0) = -1 \neq 1 = f'_+(0).$$

(2) Not defined (therefore, the value of the expression $f(0)$ cannot be calculated in the definition of the derivative).

(3) The tangent line is vertical ($f'(x) = \infty$).

A continuous function is not necessarily differentiable at every point. However, the converse is true: if a function is differentiable at a certain point, it is necessarily continuous at that point. This has been proven in Appendix 8.6.

Exercise

Derive the derivative function of $f(x) = x^3$ using the definition of the derivative (3).

Basic Exercise

Determine the equation of the tangent line to the function $f(x) = x^2 + 1$ at the point $(2, 5)$.

Basic Exercise

The task is to determine the tangent to the function $f(x) = e^{2x}$ at the point $x = 0$. This can be done as follows:

$$f'(x) = 2e^{2x},$$

$$f'(0) = 2,$$

$$f(0) = 1,$$

$$y - 1 = 2(x - 0)$$

$$y = 2x + 1.$$

Explain what the task means in terms of the graph, and justify why the solution can be obtained through the previous steps.

Exercise

Determine the equation of the tangent to the function $f(x) = x^2 + 1$ that passes through the origin.

Exercise

Show that the tangent line drawn to the exponential function cannot be parallel to the x -axis.

Notation for Derivatives

The derivative function has been denoted by f' above. Other notations that can be used are

$$\frac{df}{dx}, \quad Df \quad \text{and} \quad \dot{f}.$$

The first of these can also be written as $\frac{d}{dx}f$. The notation f' is called Lagrange's notation, the notation $\frac{df}{dx}$ is called Leibniz's notation, the notation Df is called Euler's notation, and the notation $\dot{f}$ is called Newton's notation, after their inventors. Newton's notation is less common and is not recommended for use in this course.

As with f and $f(x)$, the notations f' and $f'(x)$ are, strictly speaking, different things. The notation $f'(x)$ refers to the value of the derivative of the function f , i.e., the expression for the derivative, while f' refers to the entire derivative function, which includes the domain, codomain, and the rule that associates each element of the domain with a uniquely determined element of the codomain.

For example, for the function $f: \mathbb{R} \rightarrow \mathbb{R}$, where $f(x) = x^2$, the derivative function is $f': \mathbb{R} \rightarrow \mathbb{R}$, $f'(x) = 2x$, where $f'(x)$ refers to the expression $2x$, and f' refers to the whole structure, which includes the domain $\mathbb{R}$, the codomain $\mathbb{R}$, and the expression for the derivative $f'(x) = 2x$. As is often the case with ordinary functions, we are generally lazy in writing derivatives, presenting them in the form $f'(x) = \dots$ without mentioning the domains and ranges.

For an arbitrary value of the derivative, i.e., the expression of the derivative function, we similarly use the notations

$$f'(x), \quad \frac{d}{dx}f(x), \quad Df(x) \quad \text{and} \quad \dot{f}(x).$$

The notation $\frac{d}{dx}f(x)$ can also be written in forms such as

$$\frac{df(x)}{dx}, \quad \frac{df}{dx}(x) \quad \text{and} \quad \frac{dy}{dx},$$

where $y = f(x)$. The origin of the notation $\frac{dy}{dx}$ is as follows: If the change in the variable of the function $y = f(x)$, denoted by Δx , corresponds to the change in the value of the function denoted by Δy , then the average rate of change of the function over this interval is $\frac{\Delta y}{\Delta x}$. When the changes Δx and Δy are infinitesimally small, using the notations dx and dy , the average rate of change becomes the exact instantaneous rate of change:

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}.$$

This is an illustrative and sometimes quite useful way to think about the derivative, but precisely speaking, the notation $\frac{dy}{dx}$ is not a division operation, as discussed in Appendix 8.3.

For the value of the derivative at a specific point $x = a$, the following notations are used

$$f'(a), \quad \left. \frac{d}{dx} f(x) \right|_{x=a} \quad \text{and} \quad \dot{y}(a).$$

The notation $\left. \frac{d}{dx} f(x) \right|_{x=a}$ can also be expressed in forms such as

$$\left. \frac{df(x)}{dx} \right|_{x=a}, \quad \left. \frac{df}{dx}(x) \right|_{x=a}, \quad \text{and} \quad \left. \frac{dy}{dx} \right|_{x=a}.$$

These can also be written in enclosed forms, such as $\left(\frac{df(x)}{dx} \right) \Big|_{x=a}$. The notations

$$\frac{d}{dx} f(a), \quad \frac{df(a)}{dx}, \quad \frac{df}{dx}(a) \quad \text{and} \quad Df(a)$$

are also seen in use, but their use is not recommended in this course. This is because these notations can also be interpreted as requiring the calculation of $f(a)$ first and then taking the derivative, which would always yield zero as a result. Instead, the notations

$$\left(\frac{d}{dx} f \right) (a), \quad \left(\frac{df}{dx} \right) (a), \quad (Df)(a) \quad \text{and} \quad Df(x)|_{x=a}$$

should make sense in this context, but they are not commonly used.

What notation for derivatives should one use? Lagrange's notations f' , $f'(x)$, and $f'(a)$ are convenient. They are short, clear, and widely used. In some cases, however, Leibniz's and Euler's notations may be clearer. This is, in my opinion, particularly the case when introducing intermediate steps in differentiation. For example, compare the following notations (These intermediate steps are treated in more detail in the next chapter. For now, the only relevant thing is the notation):

$f'(x) = (2x + 3x^2)'$	$f'(x) = \frac{d}{dx}(2x + 3x^2)$	$f'(x) = D(2x + 3x^2)$
$= (2x)' + (3x^2)'$	$= \frac{d}{dx}(2x) + \frac{d}{dx}(3x^2)$	$= D(2x) + D(3x^2)$
$= 2(x)' + 3(x^2)'$	$= 2\frac{d}{dx}(x) + 3\frac{d}{dx}(x^2)$	$= 2D(x) + 3D(x^2)$
$= 2 \cdot 1 + 3 \cdot 2x$	$= 2 \cdot 1 + 3 \cdot 2x$	$= 2 \cdot 1 + 3 \cdot 2x$
$= 2 + 6x$	$= 2 + 6x$	$= 2 + 6x.$

In the intermediate steps, Lagrange's notation is concise, but it can easily be overlooked, especially in manual calculations, and there may also be situations where it is unclear with respect to which variable one is differentiating.

Leibniz's notation is convenient because it clearly emphasizes the variable with respect to which differentiation is being performed:

$$\frac{d}{dx}(x^2) = 2x, \quad \frac{d}{dx^2}(x^2) = 1 \quad \text{and} \quad \frac{d}{d(-x)}(x^2) = -2x.$$

(The last two differentiations are illustrative to understand by temporarily changing the variables to $y = x^2$ and $y = -x$.) In particular, using this notation, the chain rule for composite functions (which will be discussed in the next chapter) can be written in an illustrative form: if $y = f(x)$ and $z = g(y)$, then

$$\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx}.$$

In Euler's notation, the variable with respect to which differentiation is performed can be specified with a subscript if desired:

$$D_x(x^2) = 2x, \quad D_{x^2}(x^2) = 1 \quad \text{and} \quad D_{-x}(x^2) = -2x.$$

Calculating Derivatives in Practice

In practice, derivatives are calculated using pre-established formulas derived from the definition, rather than directly from the definition itself. These formulas are divided into two types: the derivative rules for elementary functions and more general rules that describe how to differentiate new functions formed from combinations of other functions. Let's start with the derivative formulas for elementary functions. The derivative of a power function decreases the exponent:

$$D(x^r) = rx^{r-1}, \quad r \in \mathbb{R}. \quad (4)$$

This formula applies even when the exponent is not a positive integer, allowing, for instance, the differentiation of root functions.

Example

Differentiate the following functions:

$$\begin{array}{lll} \text{(a)} f_1(x) = x^3 & \text{(b)} f_2(x) = x^\pi & \text{(c)} f_3(x) = \frac{1}{x^6} \\ \text{(d)} f_4(x) = \sqrt{x} & \text{(e)} f_5(x) = \frac{x}{\sqrt{x}}. & \end{array}$$

Solution

$$\text{(a)} \quad f_1'(x) = D(x^3) = 3x^{3-1} = 3x^2$$

$$\text{(b)} \quad f_2'(x) = D(x^\pi) = \pi x^{\pi-1}.$$

$$\text{(c)} \quad f_3'(x) = D\left(\frac{1}{x^6}\right) = D(x^{-6}) = -6x^{-6-1} = -6x^{-7} = -\frac{6}{x^7}$$

$$\text{(d)} \quad f_4'(x) = D(\sqrt{x}) = D(x^{\frac{1}{2}}) = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2x^{\frac{1}{2}}} = \frac{1}{2\sqrt{x}}$$

$$\text{(e)} \quad \frac{x}{\sqrt{x}} = \frac{x^1}{x^{\frac{1}{2}}} = x^{1-\frac{1}{2}} = x^{\frac{1}{2}},$$

so

$$f_5'(x) = D\left(\frac{x}{\sqrt{x}}\right) = D(x^{\frac{1}{2}}) = \cdots = \frac{1}{2\sqrt{x}}.$$

Basic Exercise

Differentiate the following functions:

$$(a) f_1(x) = x^{101}$$

$$(b) f_2(x) = x^{\sqrt{2}}$$

$$(c) f_3(x) = \frac{1}{x^{101}}$$

$$(d) f_4(x) = \sqrt[3]{x}$$

$$(e) f_5(x) = \frac{\sqrt[3]{x}}{\sqrt{x}}.$$

The derivatives of sine, cosine, and tangent are given by the following formulas:

$$D(\sin(x)) = \cos(x), \quad (5)$$

$$D(\cos(x)) = -\sin(x), \quad (6)$$

and

$$D(\tan(x)) = 1 + \tan^2(x) = \frac{1}{\cos^2(x)}. \quad (7)$$

The formula for differentiating tangent can be derived from the relation $\tan(x) = \sin(x)/\cos(x)$ using the quotient rule, which we will discuss later.

The differentiation formulas for trigonometric functions are valid in these forms only when the angle is in radians. This requirement extends to all formulas derived from these rules, which must also be used in radians. This is one of the main reasons why radians are often preferred over degrees in many contexts.

The derivative of the exponential function (with base e) leaves the function unchanged:

$$D(e^x) = e^x. \quad (8)$$

This is the only function (aside from the constant function $f(x) = 0$) whose derivative is the function itself. From this fact, along with the identity $a^x = (e^{\ln(a)})^x$ and the chain rule (to be introduced later), we can derive a similar formula for general exponential functions:

$$D(a^x) = \ln(a)a^x, \quad a > 0. \quad (9)$$

In this formula, the extra factor $\ln(a)$ appears, which vanishes when $a = e$.

The derivative of the natural logarithm is a power function:

$$D(\ln(x)) = \frac{1}{x}. \quad (10)$$

Note that the expression x^{-1} is not derived from any power function. Using the change-of-base formula for logarithms and the constant multiple rule for derivatives, we can derive a similar formula for general logarithmic functions:

$$D(\log_a(x)) = \frac{1}{\ln(a)} \frac{1}{x}, \quad a > 0. \quad (11)$$

In this formula, the additional factor $1/\ln(a)$ appears, which is eliminated when $a = e$. After discussing the chain rule, we will explain why the logarithmic formulas in some sources include absolute values.

Additionally, similar formulas are available for other trigonometric functions (cotangent, secant, and cosecant), the inverse trigonometric functions, and the hyperbolic functions and their inverses.

General Differentiation Rules

The previous formulas do not explain how to differentiate functions that are not in standard forms, such as $f(x) = 3x^3$. For such cases, we need the following formulas, which can be derived from the definition of a derivative. **These rules should be memorized to make problem-solving meaningful.** If you have not memorized them, it likely means you haven't practiced enough problems.

The derivative of a constant function is zero:

$$D(c) = 0. \quad (12)$$

In fact, a constant function is the only function with a derivative of zero. This rule becomes evident when considering the function's graph.

A constant factor c can be pulled outside the derivative:

$$D(cf(x)) = cD(f(x)) \quad (13)$$

This property is also clear when thinking about how a constant factor changes the values of the function, and therefore its rate of change. Using this rule, the derivative of the function $f(x) = 3x^3$ can be calculated as follows:

$$D(3x^3) = 3D(x^3) = 3 \cdot 3x^2 = 9x^2.$$

Generally, as you become more familiar with differentiation, this intermediate step is often skipped, but while learning, it is useful to write out all steps.

For the sum or difference of functions, each term can be differentiated individually:

$$D(f(x) \pm g(x)) = D(f(x)) \pm D(g(x)). \quad (14)$$

This property also makes sense when visualized through graphs.

Basic Exercise

Which differentiation rules are used in each step of the following calculation?

$$\begin{aligned} D(5x^3 + 4x^2 + x + 2) &= D(5x^3) + D(4x^2) + D(x) + D(2) \\ &= 5D(x^3) + 4D(x^2) + 1 + 0 \\ &= 5 \cdot 3x^2 + 4 \cdot 2x + 1 \\ &= 15x^2 + 8x + 1 \end{aligned}$$

Once you become more comfortable with differentiation, you may omit the second and third steps, but until then, it's helpful to include all intermediate steps.

For the product of two functions, the following rule applies:

$$D(f(x)g(x)) = D(f(x))g(x) + f(x)D(g(x)) \quad (15)$$

Note that it does not hold that $D(f(x)g(x)) = D(f(x))D(g(x))$. The reason the product rule works as it does can be understood visually by considering areas, as discussed in Appendix 8.5.

Basic Exercise

Compute

$$D(4e^x \cos(x)).$$

Basic Exercise

Calculate x^2 using the product rule.

If differentiating a product of more than two functions, the rule above can still be applied by using it multiple times. Alternatively, the rule can be extended to products of multiple functions. For instance, we have

$$D(f(x)g(x)h(x)) = D(f(x))g(x)h(x) + f(x)Dg(x)h(x) + f(x)g(x)Dh(x). \quad (16)$$

The product rule for multiple functions operates similarly: for a product of n functions, there will be n terms, each with only one function differentiated.

Example

To differentiate $f(x) = 2^x x^2 e^x$, apply the product rule (15) twice:

$$\begin{aligned} D(2^x x^2 e^x) &= D(2^x (x^2 e^x)) = D(2^x)(x^2 e^x) + 2^x D(x^2 e^x) \\ &= \ln(2) 2^x (x^2 e^x) + 2^x (2x e^x + x^2 e^x) \\ &= (\ln(2)x^2 + 2x + x^2) 2^x e^x \\ &= ((\ln(2) + 1)x + 2) 2^x x e^x. \end{aligned}$$

Alternatively, using the formula for the product of three functions directly,

$$D(2^x x^2 e^x) = \ln(2) 2^x x^2 e^x + 2 \cdot 2^x x e^x + 2^x x^2 e^x = ((\ln(2) + 1)x + 2) 2^x x e^x.$$

Exercise

Derive formula (16).

The derivative of the quotient of functions is given by

$$D\left(\frac{f(x)}{g(x)}\right) = \frac{D(f(x))g(x) - f(x)D(g(x))}{(g(x))^2}. \quad (17)$$

This can be derived straightforwardly from the product rule and the derivative of a composite function that will be presented next. In deriving the formula, it also becomes evident which term in the numerator should have the minus sign, which can be difficult to remember. It should also be noted that generally, it is not advisable to expand the power of the denominator.

Example

Let's calculate part (e) on page 154 using the previous formula:

$$D\left(\frac{x}{\sqrt{x}}\right) = \frac{D(x)\sqrt{x} - xD(\sqrt{x})}{\sqrt{x}^2} = \frac{1 \cdot \sqrt{x} - x \cdot \frac{1}{2\sqrt{x}}}{\sqrt{x}^2} = \frac{\sqrt{x} - \frac{\sqrt{x}}{2}}{x} = \frac{\left(\frac{\sqrt{x}}{2}\right)}{x} = \frac{1}{2\sqrt{x}}.$$

Thus, we obtained the same answer, but this technique was considerably more labor-intensive, and we also had to calculate the derivative $D(\sqrt{x})$, which was practically the entire solution in the previous method.

Basic Exercise

Calculate the derivatives of the following functions using both the quotient rule and by simplifying their expressions first, and then using the power function derivative rule:

$$(a) f(x) = \frac{2}{x^3}$$

$$(b) g(t) = \frac{t^5 + t^3 + t}{t^3}.$$

Basic Exercise

Differentiate the following functions:

$$(a) f(x) = \frac{\sin(x)}{e^x}$$

$$(b) g(x) = \frac{3x^2\sqrt{x}x^{-3}}{4\frac{x^5}{\sqrt{x}}x^{-2}}.$$

Composite functions are differentiated using the formula:

$$D(g(f(x))) = g'(f(x))D(f(x)). \quad (18)$$

This formula, known as the **chain rule**, is needed continuously, and to use this formula, one must be able to recognize which nested functions make up the function. When applying the formula, one must:

- (1) Identify the outer function g .
- (2) Differentiate the outer function.
- (3) Substitute the expression of the inner function $f(x)$ for the variable in the derivative of the outer function g' .
- (4) Multiply the result by the derivative of the inner function $f'(x)$.

If the inner function is itself a composite function, the chain rule must be applied again to calculate its derivative, as shown in part (c) of the following example.

Example

Differentiate the following functions:

$$(a) f_1 = 3(2x + 3)^{2020}$$

$$(b) f_2(x) = \ln(-x)$$

$$(c) f_3(x) = \cos^2(e^{2x+3})$$

$$(d) f_4(x) = \cos^2(e^{2x+3} + 4x).$$

Solution

- (a) Here, the outer function is $u(x) = 3x^{2020}$, and its derivative is $u'(x) = 6060x^{2019}$, with the inner function being $s(x) = 2x + 3$. Thus, we have:

$$\begin{aligned} f_1'(x) &= 6060(2x + 3)^{2019} D(2x + 3) = 6060(2x + 3)^{2019} \cdot 2 \\ &= 12120(2x + 3)^{2019} \end{aligned}$$

(b)

$$f_2'(x) = \frac{1}{-x} D(-x) = \frac{-1}{-x} = \frac{1}{x}$$

Solution (continued)

(c)

$$\begin{aligned}f_3'(x) &= 2 \cos(e^{2x+3}) \cdot D(\cos(e^{2x+3})) \\&= 2 \cos(e^{2x+3}) \cdot (-\sin(e^{2x+3}) \cdot D(e^{2x+3})) \\&= 2 \cos(e^{2x+3}) \cdot (-\sin(e^{2x+3}) \cdot (e^{2x+3} \cdot D(2x+3))) \\&= 2 \cos(e^{2x+3}) \cdot (-\sin(e^{2x+3}) \cdot (e^{2x+3} \cdot 2)) \\&= -4 \cos(e^{2x+3}) \sin(e^{2x+3}) e^{2x+3}\end{aligned}$$

Here, the logic is as follows: First, we see that the outermost function is raised to the power of two and its inner function is $\cos(e^{2x+3})$.

Therefore, we first differentiate the square function, place the inner function $\cos(e^{2x+3})$ in the place of the variable in the derivative obtained, and multiply it by the derivative of the inner function. After that, we simply need to calculate the derivative of the inner function $D(\cos(e^{2x+3}))$, which follows the same logic as above. We examine the outer function and its inner function, and so on.

Solution (continued)

(d)

$$f_4'(x) = 2 \cos(e^{2x+3} + 4x) \sin(e^{2x+3} + 4x)(2e^{2x+3} + 4)$$

Note that the inner function of the function e^{2x+3} is not $f(x) = e$, but rather the inner function is $s(x) = 2x + 3$ and the outer function is $u(x) = e^x$. If this is somewhat confusing, rewriting the expression as $\exp(2x + 3)$ and comparing it, for example, to the expression $\sin(2x + 3)$ should clarify the situation. If the composite function contains a variable, its inner function cannot be an expression that does not include the variable x . If the inner function were $f(x) = e$, then the composite function could not retain any variable. For example, in the case of the outer function $g(x) = 2x + 3$, it would hold that

$$g(f(x)) = g(e) = 2e + 3.$$

The inner function of $(2x + 3)^e$ is $s(x) = 2x + 3$ and the outer function is $u(x) = x^e$.

Basic Exercise

Differentiate the following functions:

$$(a) f(x) = 2(x^3 + x + 4)^5$$

$$(b) g(x) = e^{\cos(x)+x^2}$$

$$(c) h(x) = (2x - 3)^4 \sin(5x).$$

Basic Exercise

Compute the following derivatives:

$$D \sin (2(3x^2 - 1)^4 + 5x)$$

and

$$D \sin (2(3x^2 - 1)^4 + 5x).$$

The derivative of the function

$$\ln |x| = \begin{cases} \ln(x), & x > 0 \\ \ln(-x), & x < 0 \end{cases}$$

satisfies

$$D(\ln |x|) = \frac{1}{x}.$$

For example, in MAOL, this formulation is used, which thus also includes the case $x < 0$ in the differentiation formula. The formulation of the formula using the absolute value helps to understand the corresponding integration formula, which requires the absolute value. In addition, the derivative of the natural logarithm is $1/x$ in many other cases, as seen in the following exercise.

Basic Exercise

Differentiate the following expressions:

(a) $\ln(2x)$

(b) $\ln(3x)$

(c) $\ln(ax), \quad a > 0.$

Basic Exercise

Differentiate the function

$$f(x) = 2x^2 - \frac{2}{3} + \ln(2x) - \frac{4}{x^3} + 5\sqrt{x} + \frac{\sin(x)}{\cos(x)} + x^2 e^{2x}.$$

Exercise

Differentiate the following expressions:

$$(a) (e^{\ln(a)})^x, a > 0 \quad (b) \frac{\ln(x)}{\ln(a)}, a > 0 \quad (c) f(x)(g(x))^{-1}.$$

What did you calculate in this task?

Finally, it should be noted that the presented formulas can also be written using other notations for derivatives. For example, it holds that

$$\frac{d}{dx}(cf(x)) = c \frac{d}{dx}f(x) \quad \text{and} \quad (f(x)g(x))' = f'(x)g(x) + f(x)g'(x).$$

In particular, when we denote $f(x) = y$ and $g(y) = z$, the derivative formula for the composite function is given by

$$\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx}.$$

For example, the differentiation of the function $g(x) = \ln(-x)$ can also be thought of as

$$\frac{dg}{dx} = \frac{d \ln(-x)}{d(-x)} \frac{d(-x)}{dx} = \frac{1}{-x}(-1) = \frac{1}{x}.$$

Derivative of the Inverse Function

We will introduce this topic by deriving the differentiation formula for the natural logarithm (assuming the differentiation formula for the exponential function is known).

When $x > 0$, it is known that

$$e^{\ln(x)} = x.$$

Since both sides represent the same function, their derivatives must also be the same, so we have

$$D(e^{\ln(x)}) = D(x)$$

or equivalently,

$$e^{\ln(x)} D(\ln(x)) = 1.$$

Thus,

$$D(\ln(x)) = \frac{1}{e^{\ln(x)}}$$

which simplifies to

$$D(\ln(x)) = \frac{1}{x}.$$

To clarify, partial differentiation only holds the equation true if both sides are equal for **every value of the variable**. As a counterexample, you might try solving the equation $x^2 = 4$ as well as the equation obtained by partially differentiating this.

Basic Exercise

Assuming the power rule for differentiation is known, derive the differentiation formula for the square root function using the previous technique.

Basic Exercise

Assuming the differentiation rule for sine is known, derive the differentiation formula for the inverse sine function $\arcsin$.

With the previous technique, we can derive the general rule for differentiating an inverse function. If a function f has an inverse function, and we write $f(x) = y$, then

$$\begin{aligned}f(f^{-1}(y)) &= y \\ \frac{d}{dy}f(f^{-1}(y)) &= \frac{d}{dy}y \\ f'(f^{-1}(y))\frac{d}{dy}f^{-1}(y) &= 1 \\ \frac{d}{dy}f^{-1}(y) &= \frac{1}{f'(f^{-1}(y))} \\ \frac{d}{dy}f^{-1}(y) &= \frac{1}{f'(x)}.\end{aligned}$$

The form of the formula

$$(f^{-1})'(y) = \frac{1}{f'(x)}, \quad \text{where } y = f(x), \quad (19)$$

can sometimes be used even when the inverse function is unknown.

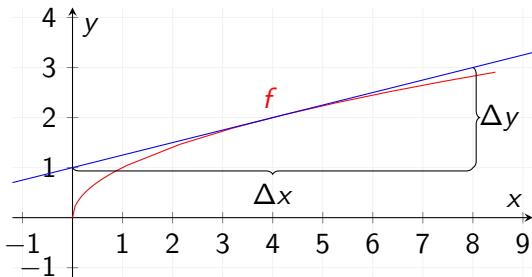
Example

The function $f(x) = x^5 + x^3 + x + 1$ has an inverse function since it is strictly increasing, though we cannot solve for it explicitly. However, we can find the value of the derivative $(f^{-1})'(1)$. Since $f(0) = 1$, we have

$$(f^{-1})'(1) = \frac{1}{f'(0)} = \frac{1}{5 \cdot 0^4 + 3 \cdot 0^2 + 1} = \frac{1}{1} = 1.$$

Thus, the value of the derivative of the inverse function is the reciprocal of the derivative of the original function. Alternatively, this can be formulated as: $(f^{-1})'(y)f'(x) = 1$, which is easy to remember.

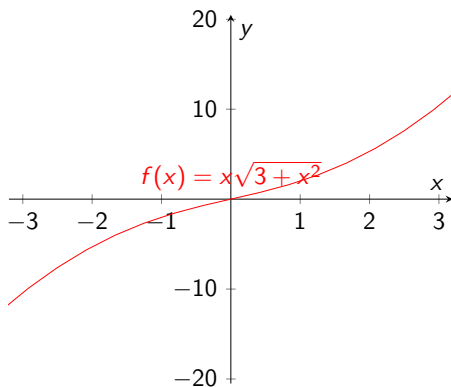
Interpreting the relationship between derivatives in terms of tangent lines makes the connection quite intuitive.



Example

Let $f(x) = x\sqrt{3+x^2}$. Compute $(f^{-1})'(-2)$.

Solution



The goal is to determine the value of x corresponding to $f(x) = -2$, calculate the derivative of the function f at this point, and then take the reciprocal of this result.

Solution (continued)

x :

$$f(x) = -2$$

$$x\sqrt{3+x^2} = -2$$

$$|(\)|^2, x\sqrt{3+x^2} < 0$$

$$x^2(3+x^2) = 4$$

$$x^4 + 3x^2 - 4 = 0$$

$$(x^2)^2 + 3x^2 - 4 = 0$$

$$x^2 = \frac{-3 \pm \sqrt{3^2 - 4 \cdot 1 \cdot (-4)}}{2 \cdot 1}$$

$$= \frac{-3 \pm 5}{2}$$

$$x^2 = -4 \quad \text{or} \quad x^2 = 1$$

$$\text{no solution} \quad \text{or} \quad x = \pm 1.$$

The candidates $x = \pm 1$ are not necessarily solutions, as they must satisfy the conditions $3 + x^2 \geq 0$ (Why?) and $x\sqrt{3+x^2} < 0$ (Why?). Only $x = -1$ satisfies both conditions, so this is the required value of x (This could also have been checked by calculating $f(-1)$ and $f(1)$).

Solution (continued)

Derivative:

$$\begin{aligned}f'(x) &= \frac{d}{dx}(x) \cdot \sqrt{3+x^2} + x \cdot \frac{d}{dx}(\sqrt{3+x^2}) \\&= 1 \cdot \sqrt{3+x^2} + x \cdot \frac{1}{2}(3+x^2)^{-\frac{1}{2}} \cdot 2x \\&= \sqrt{3+x^2} + \frac{x^2}{\sqrt{3+x^2}},\end{aligned}$$

$$f'(-1) = \sqrt{3+(-1)^2} + \frac{(-1)^2}{\sqrt{3+(-1)^2}} = \frac{5}{2}.$$

Derivative of the inverse function:

$$(f^{-1})'(-2) = \frac{1}{f'(-1)} = \frac{1}{\left(\frac{5}{2}\right)} = \cdots = \frac{2}{5}.$$

Basic Exercise

Let $f(x) = \frac{x}{x+1}$. Calculate $(f^{-1})'(\frac{1}{2})$.

Finally, we list the derivative formulas for the inverse functions of sine, cosine, and tangent:

$$(1) D \arcsin(x) = \frac{1}{\sqrt{1-x^2}}$$

$$(2) D \arccos(x) = -\frac{1}{\sqrt{1-x^2}}$$

$$(3) D \arctan(x) = \frac{1}{x^2+1}.$$

These derivatives are seldom needed directly, though they appear somewhat more frequently in integration. For example, we have

$$\int \frac{2}{x^2+1} dx = 2 \arctan(x) + C.$$

Higher Order Derivatives

When the function f is differentiated, a new function f' is obtained. This function can be differentiated again, resulting in the function $(f')' = f''$. This is called the **second derivative** of the function f .

The derivative of a function describes the instantaneous rate of change of the original function at each point. The second derivative, on the other hand, describes the instantaneous rate of change of the derivative function at each point.

Example

The derivative of the function $f(x) = x^2$ is $f'(x) = 2x$. Its second derivative is $f''(x) = 2$. The value of the derivative at the point $x = 1$ is $f'(1) = 2 \cdot 1 = 2$. The value of the second derivative is everywhere 2, indicating that the value of the derivative increases by two whenever the value of the variable increases by one. Based on this, it can be concluded that it should hold, for example, $f'(2) = 4$ and $f'(0) = 0$. Calculating can confirm that this is indeed the case.

Example

The derivative of the function $f(x) = x^3$ is $f'(x) = 3x^2$. Its second derivative is $f''(x) = 6x$. The derivative at the point $x = 1$ is $f'(1) = 3 \cdot 1^2 = 3$. The value of the second derivative at the same point is $f''(1) = 6$. This value indicates that the derivative's value increases **instantaneously** at a rate of 6 when moving from that point in the direction of the positive x -axis. However, since the value of the second derivative is not constant, accurate conclusions cannot be made about the values of the function's derivative at, for example, the points $x = 2$ and $x = 0$. It holds that $f'(2) = 12$ and $f'(0) = 3$, so the derivative increases, on average, faster than the value 6 on the interval $[1, 2]$ and slower than the value 3 on the interval $[0, 1]$.

The second derivative can be differentiated again, resulting in the third derivative $(f'')' = f'''$. This can be differentiated further to obtain the fourth derivative f'''' and so on. Generally, starting from the third derivative, the notation $f^{(n)}$ is used instead of the factorial notation, as it indicates more clearly which derivative is being referred to. The number n is called the **order of the derivative**, and the n -th derivative is also referred to as the **n -th order derivative**. In particular, the function itself can be interpreted as the zeroth order derivative and denoted by $f^{(0)} = f$. This allows certain formulas to be expressed in a general form, which comes up when discussing Taylor polynomials.

Higher order derivatives can also be denoted by:

$$\frac{d}{dx} \left(\frac{d}{dx} \left(\cdots \frac{df}{dx} \right) \right) = \frac{d^n f}{dx^n}$$

and

$$D(D(\cdots Df)) = D^n f.$$

For the value of the derivative at the point $x = a$, the following notations are used:

$$\left. \frac{d^n f}{dx^n} \right|_{x=a} \quad \text{and} \quad D^n f(x)|_{x=a}.$$

The interpretation of higher order derivatives is similar to that of the first and second order cases, meaning the n -th order derivative describes the rate of change of the $(n - 1)$ -th order derivative. However, in general, the first two derivatives are the most important. They allow for conclusions to be drawn about the essential characteristics of the function's graph and to investigate the function's local minima and maxima. In particular, if the function in question describes the position of an object, the first derivative describes the object's velocity and the second derivative describes its acceleration.

Basic Exercise

- (a) Calculate the first five derivatives of the function $f(x) = 2 + 3x + 4x^2 + e^x + \cos(2x)$.
- (b) Calculate the first three derivatives of the function $f(x) = e^x x^2$.

Extreme Values of a Function

Terms Related to Extreme Values

Let's start by reviewing terms related to extreme values. There are two types of extreme values: local extreme values and global (or absolute) extreme values. Let's examine their interpretations using the following figure.

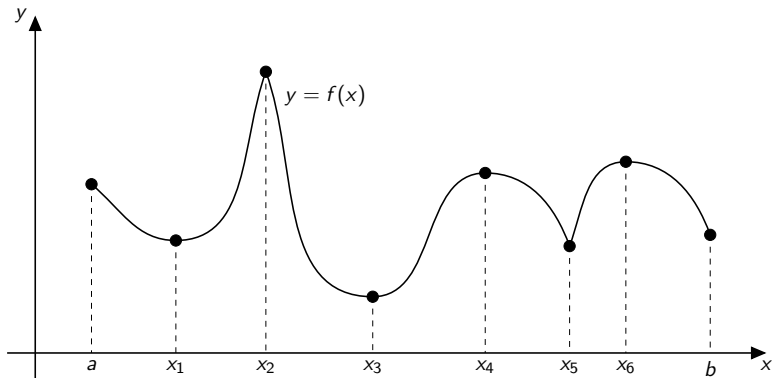


Figure: A graph with different extrema points. $D(f) = [a, b]$.

In the previous figure,

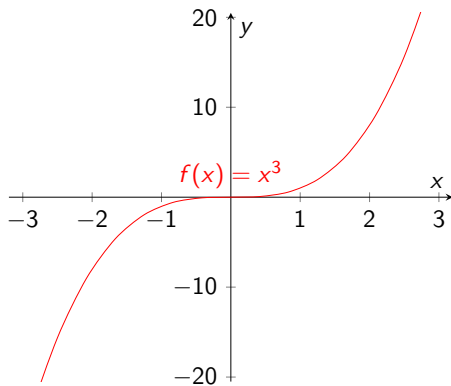
- the local maxima (or maximum points) are a , x_2 , x_4 , and x_6
- the local maximum values are $f(a)$, $f(x_2)$, $f(x_4)$, and $f(x_6)$
- the local minima points are x_1 , x_3 , x_5 , and b
- the local minimum values are $f(x_1)$, $f(x_3)$, $f(x_5)$, and $f(b)$
- the local extreme points are a , x_1 , x_2 , x_3 , x_4 , x_5 , x_6 , and b
- the local extreme values are $f(a)$, $f(x_1)$, $f(x_2)$, $f(x_3)$, $f(x_4)$, $f(x_5)$, $f(x_6)$, and $f(b)$
- the absolute (or global) maxima points are x_2
- the absolute maximum values are $f(x_2)$
- the absolute minima points are x_3
- the absolute minimum values are $f(x_3)$
- the absolute extreme points are x_2 and x_3
- the absolute extreme values are $f(x_2)$ and $f(x_3)$.

Absolute extreme values refer to the largest and smallest values of the function. There can also be multiple absolute minimum or maximum points, such as with the sine function. Local extreme values, on the other hand, refer to the largest or smallest value of the function over some smaller interval.

Formally, x_0 is a local extreme point if for some open interval centered at x_0 , $A = (x_0 - \delta, x_0 + \delta)$, where $\delta > 0$, it holds that $f(x_0) \geq f(x)$ or $f(x_0) \leq f(x)$ for every $x \in A$.

Extreme Values of a Continuous Function on a Closed Interval. The derivative is related to local extreme points. The slope of the tangent line drawn to the graph of the function is negative when the graph is decreasing and positive when the graph is increasing. When the slope is zero, the graph is (temporarily) horizontal. At such a point, i.e., at a point where the derivative is zero, there is usually a local maximum or minimum, as seen at the points x_1 , x_3 , x_4 , and x_6 in the previous figure.

However, this is not necessarily the case, as illustrated by the function $f(x) = x^3$ at the point $x = 0$.



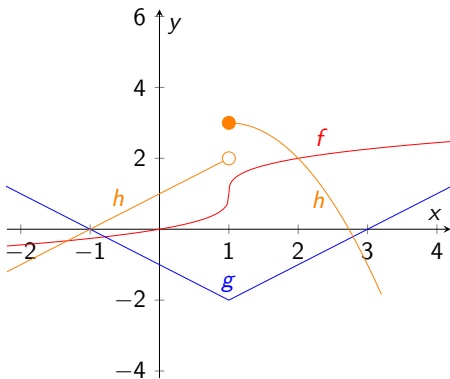
A zero of the derivative that is not a local extreme point is called a **saddle point**.

In summary, if the derivative of a function is zero, then the function has either a local extreme point or a saddle point at that location.

The endpoints of the closed interval can also be local extreme points of the function. Additionally, we note that local extreme values can occur at points x_2 and x_5 , where the graph of the function has a peak. In such points, the derivative of the function does not exist. These points are referred to as **singular points** of the function.

Another type of singular points are those where the derivative approaches infinity or negative infinity, and where the function is defined, but the graph exhibits a jump.

Points where the function is not defined, such as $x = 0$ in the case of the function $f(x) = 1/x$, are not considered singular points. The accompanying figure illustrates various functions with a singular point at $x = 1$.



When plotting the graphs of various continuous functions, one quickly observes that besides the previously discussed cases, there are no other points where local extrema could exist.

Theorem

A **continuous** function on a **closed** interval $[a, b]$ can have extreme values only at the following points:

- the endpoints of the interval a and b
- the zeros of the derivative
- the singular points (where the derivative is not defined).

The type of the zeros of the derivative—whether it is a local minimum, local maximum, or a saddle point—can be determined based on the sign chart of the derivative, which will be discussed in the following example. The types of the endpoint values and singular values can also be examined in a similar manner.

Example

Determine the local extreme points and their types for the functions

$$(a) f(x) = x^4 - 6x^2 - 8x$$

$$(b) f(x) = \begin{cases} x + 1, & x < 1 \\ x^2 - 4x + 5, & x \geq 1 \end{cases}$$

on the interval $[-5, 5]$.

Solution

- (a) Since the function is differentiable everywhere and the interval under consideration is closed, the local extreme points can be found at the endpoints of the interval or at the zeros of the derivative.

Solution (continued)

The derivative of the function is

$f'(x) = 4x^3 - 12x - 8 = 4(x^3 - 3x - 2)$. We determine its zeros:

$$4(x^3 - 3x - 2) = 0$$

$$x^3 - 3x - 2 = 0$$

$$(x + 1)(x^2 + x - 2) = 0$$

$$(x + 1)(x + 1)(x - 2) = 0$$

$$x = -1 \quad \text{or} \quad x = 2.$$

To determine the type of local extreme value, it is sufficient to know whether the function's graph is increasing or decreasing on either side of the zeros. This can be inferred from the sign of the derivative, so we will create a sign chart for the derivative.

Solution (continued)

To do this, we need to determine the sign of the derivative in the intervals $(-\infty, -1)$, $(-1, 2)$, and $(2, \infty)$ (that is, practically solve the inequality $f'(x) > 0$):

$$f'(-3) = 4((-3)^3 - 3(-3) - 2) = -80 < 0,$$

thus $f'(x) < 0$ in the interval $(-\infty, -1)$.

$$f'(0) = 4(0^3 - 3 \cdot 0 - 2) = -8 < 0,$$

thus $f'(x) < 0$ in the interval $(-1, 2)$.

$$f'(3) = 4(3^3 - 3 \cdot 3 - 2) = 64 > 0,$$

thus $f'(x) > 0$ in the interval $(2, \infty)$. Based on this information, we can construct the following sign chart for the derivative.

	-5	-1	2	5	x		
$f'(x)$		-		-		+	

Solution (continued)

Furthermore, we can add the **flow chart** of the function, which indicates whether the graph of the function is increasing or decreasing in the examined intervals.

	-5	-1	2	5	x
$f'(x)$	-	-	+		
$f(x)$	$\searrow$	$\searrow$	$\nearrow$		

Based on the flow chart, it is clear that the function f has local maxima at points $x = -5$ and $x = 5$, as well as a local minimum at point $x = 2$. (At point $x = -1$, there is a saddle point.)

Solution (continued)

(b) Let's start by verifying that the function is continuous at point $x = 1$.

$$\lim_{x \rightarrow 1^-} f(x) = 1 + 1 = 2,$$

$$f(1) = 1^2 - 4 \cdot 1 + 5 = 2$$

and

$$\lim_{x \rightarrow 1^+} f(x) = 1^2 - 4 \cdot 1 + 5 = 2,$$

so the condition $\lim_{x \rightarrow 1^-} f(x) = f(1) = \lim_{x \rightarrow 1^+} f(x)$ holds. Thus, f is continuous at point $x = 1$ and throughout the examined interval.

Now, the function may not necessarily be differentiable at point $x = 1$, so it could have extrema not only at the endpoints and the zeros of the derivative but also at this point. Determining whether it is a singular point, a zero of the derivative, or neither is cumbersome, but ascertaining whether it is a local extremum can be straightforwardly done using the flow chart.

Solution (continued)

Zeros of derivative: In the interval $(-5, 1)$, the derivative is $f'(x) = 1$, so it has no zeros. In the interval $(1, 5)$, we have:

$$f'(x) = 2x - 4,$$

$$f'(x) = 0$$

$$2x - 4 = 0$$

$$x = 2,$$

$$f'(1.5) = 2 \cdot 1.5 - 4 = -1 < 0$$

and

$$f'(3) = 2 \cdot 3 - 4 = 2 > 0.$$

Solution (continued)

Thus, we obtain the following flow chart.

	-5	1	2	5	x
$f'(x)$	+	-	+		
$f(x)$	$\nearrow$	$\searrow$	$\nearrow$		

Thus, the function f has local maxima at points $x = 1$ and $x = 5$, as well as local minima at points $x = -5$ and $x = 2$.

The minimum and maximum values of a continuous function can be found by determining the local extrema and selecting the smallest and largest values among them. These values always exist.

Theorem

A continuous function has a minimum and maximum value on a closed interval.

If a function is not only continuous on a closed interval but also differentiable, then the minimum and maximum values can be found at either the endpoints or the critical points of the derivative.

Example

Determine the maximum and minimum values of the function

$$(a) f(x) = x^4 - 6x^2 - 8x$$

$$(b) g(x) = \begin{cases} x + 1, & x < 1 \\ x^2 - 4x + 5, & x \geq 1 \end{cases}$$

on the interval $[-5, 5]$.

Solution

- (a) In this case, we could use the flow chart as in the previous example, but this is easier: Since f is a polynomial and thus differentiable everywhere, the maximum and minimum values are obtained either at the endpoints or the zeros of the derivative. The zeros are $x = -1$ and $x = 2$ (as in the previous example). The values at these points are $f(-1) = 3$ and $f(2) = -24$. The values at the endpoints are $f(-5) = 515$ and $f(5) = 435$. Therefore, the minimum value of the function on this interval is $f_{\min} = f(2) = -24$, and the maximum value is $f_{\max} = f(-5) = 515$.

Solution (continued)

- (b) Similarly, we could use the flow chart as in the previous example, but this is easier: Since g is differentiable everywhere except possibly at $x = 1$, the maximum and minimum values can be obtained either at the endpoints, the zeros of the derivative, or at point $x = 1$, which could be a zero of the derivative, a singular point, or neither.

However, there is no need to determine what type of point $x = 1$ is; we can simply calculate its value and see if it yields a maximum or minimum value. (Based on the graph, it is a singular point.)

Values at the endpoints: $g(-5) = -5 + 1 = -4$,

$$g(5) = 5^2 - 4 \cdot 5 + 5 = 10.$$

Zeros of derivative: $x = 2$ (which can be determined as in the previous task), $g(2) = 2^2 - 4 \cdot 2 + 5 = 1$.

Possible singular point/zero of derivative: $g(1) = 2$.

Thus, the minimum and maximum values are found at the endpoints:

$$g_{\min} = g(-5) = -4, \quad g_{\max} = g(5) = 10.$$

Basic Exercise

Determine the maximum and minimum values of the function $f(x) = 3x^5 - 5x^3$ on the interval $[-1, 2]$.

Exercise

Determine the local maximum and minimum values of the function $p(x) = x^{2/3} + (x - 1)^{2/3}$ on the interval $[-1, 2]$.

Example

Let $f(x) = 2\sin(x) + \cos(x)$.

- (a) Find all zeros of derivative and singular points of the function f .
- (b) Determine the minimum and maximum points of the function f on the interval $[0, \frac{2\pi}{3}]$.

Solution

(a)

$$f(x) = 2 \sin(x) + \cos(x)$$

$$f'(x) = 2 \cos(x) - \sin(x)$$

Since f' is defined everywhere, the function f has no singular points.

Points where the derivative is zero:

$$f'(x) = 0$$

$$2 \cos(x) - \sin(x) = 0$$

$$\frac{\sin(x)}{\cos(x)} = 2$$

$$\tan(x) = 2$$

$$x = \tan^{-1}(2) + \pi n, \quad n \in \mathbb{Z}$$

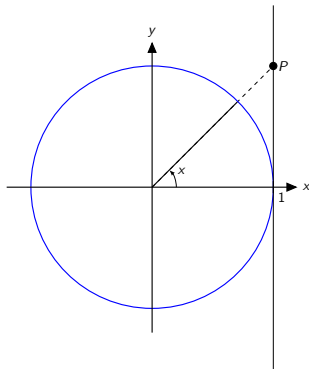


Figure: $P = (1, \tan(x))$

Solution (continued)

(b) The interval of consideration is $[0, \frac{2\pi}{3}]$, so there is only $x = \tan^{-1}(2)$ among the zeros of derivative in this interval. The extrema can now be found either at the endpoints of the interval or at the zeros of the derivative (Why?), so it would suffice to calculate the function values at these points and select the maximum and minimum among them. However, for the sake of illustration, let's examine this with a flow chart. First, let's calculate a couple of derivative values:

$$f' \left(\frac{\pi}{4} \right) = 2 \cos \left(\frac{\pi}{4} \right) - \sin \left(\frac{\pi}{4} \right) = 2 \cdot \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} > 0$$

$$f' \left(\frac{\pi}{2} \right) = 2 \cos \left(\frac{\pi}{2} \right) - \sin \left(\frac{\pi}{2} \right) = 2 \cdot 0 - 1 = -1 < 0.$$

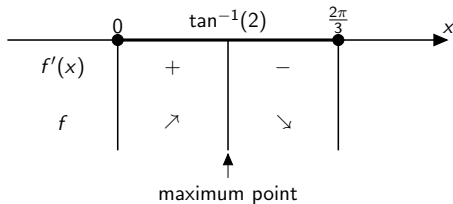


Figure: First derivative sign chart.

Solution (continued)

According to the flow chart, $x = \tan^{-1}(2)$ is a maximum point. To find the minimum point, we need to calculate the function values at the endpoints of the interval:

$$\begin{aligned} f(0) &= 2 \cdot \sin(0) + \cos(0) = 1 \\ f\left(\frac{2\pi}{3}\right) &= 2 \cdot \sin\left(\frac{2\pi}{3}\right) + \cos\left(\frac{2\pi}{3}\right) \\ &= 2 \cdot \left(\frac{\sqrt{3}}{2}\right) + \left(-\frac{1}{2}\right) = \sqrt{3} - \frac{1}{2} > 1. \end{aligned}$$

Thus, $x = 0$ is a minimum point.

Extrema on Open Intervals or When the Function is Not Continuous

A continuous function attains a maximum and minimum value on a closed interval. If the interval is not closed or the function is not continuous, this may not be the case, as illustrated in the following example.

Example

Determine the absolute extrema of the function $f(x) = \frac{1}{x^2}$ on the intervals

(a) $[1, 2]$

(b) $(1, 2]$

(c) $[-1, 1]$.

Solution

(a)

$$f(x) = \frac{1}{x^2} = x^{-2}, \quad x \neq 0,$$

$$f'(x) = -2x^{-3} = -\frac{2}{x^3},$$

so the derivative is positive when $x < 0$ and negative when $x > 0$.

Thus, f is decreasing on the interval considered. Therefore, the absolute minimum on the interval $[1, 2]$ is at $x = 2$, and the absolute maximum is at $x = 1$.

Solution (continued)

- (b) On the interval $(1, 2]$, the minimum point remains $x = 2$, but now there is no maximum point, as when approaching $x = 1$ from the right, larger and larger values are obtained, but the point itself is never reached. (If the absence of a maximum point seems confusing, try to articulate the point at which the largest value would be attained. No matter what you say, there will always be a number between your given value and 1, and at that point, a larger value will be obtained.)
- (c) Now, it should be noted that at $x = 0$ there is division by zero, so f is not continuous at this point. Let's investigate how the function values behave as we approach zero from either side:

$$\lim_{x \rightarrow 0^{\pm}} \underbrace{\frac{1}{x^2}}_{\rightarrow 0^+} = \infty.$$

Thus, the function does not have a maximum value on the interval. This could also have been concluded from the flow chart (Try it).

Based on part (a), f is increasing when $x < 0$ and decreasing when $x > 0$, so the minimum points are at $x = -1$ or $x = 1$. Since $f(-1) = 1 = f(1)$, both are minimum points.

In the previous example, division by zero led to the conclusion that the function cannot have a maximum value on the interval containing that point. This does not always happen. For example, consider the case

$$f(x) = -\frac{x^2(x+1)}{x+1}.$$

On the other hand, division by zero may also mean that the function cannot have a minimum value or a maximum value on the interval containing that point. For example, consider the case

$$f(x) = \frac{1}{x}.$$

In summary, we can state the following.

Theorem

On an open interval (a, b) or half-open interval $(a, b]$ or $[a, b)$, a continuous function can attain extrema only at the following points:

- an endpoint **included** in the interval
- the zeros of the derivative
- the singular points (where the derivative is not defined).

Absolute extrema may not necessarily exist on the interval.

Therefore, a continuous function on the considered interval does not have absolute extrema if local extrema yield values that are lower/higher when approaching an open endpoint. This can be investigated by calculating the limits corresponding to the endpoints. A flow chart often clarifies the situation.

Example

Determine the maximum and minimum values of the function

$$f: (2, 8) \rightarrow \mathbb{R}, f(x) = \begin{cases} x^2 - 1, & x \in (-2, 2] \\ 6 - x, & x \in (2, 8) \end{cases}.$$

Solution

Let's illustrate the situation with a flow chart.

The function may be discontinuous at $x = 2$.

Zero of the derivative: In the case $x < 2$, the derivative is $f'(x) = 2x$ and it has only one zero at $x = 0$. In the case $x > 2$, the derivative is $f'(x) = -1$, and it has no zeros.

Solution (continued)

Flow chart:

$$f'(-1) = 2(-1) = -2 < 0,$$

$$f'(1) = 2 \cdot 1 = 2 > 0,$$

$$f'(6) = -1 < 0,$$

	-2	0	2	8	x
$f'(x)$	-	+	-		
$f(x)$	$\searrow$	$\nearrow$	$\searrow$		

The point $x = 0$ is thus an absolute minimum unless the function takes on a smaller value as $x \rightarrow 8^-$. Let's clarify this by calculating:

$$f(0) = 0^2 - 1 = -1$$

and

$$\lim_{x \rightarrow 8^-} f(x) = f(8) = 6 - 8 = -2,$$

so there is no absolute minimum.

Solution (continued)

The point $x = 2$ could be a maximum unless greater values are obtained as $x \rightarrow -2^-$ and $x \rightarrow 2^+$. Let's investigate by calculating:

$$\lim_{x \rightarrow -2^-} f(x) = f(-2) = (-2)^2 - 1 = 3,$$

$$f(2) = 2^2 - 1 = 3,$$

and

$$\lim_{x \rightarrow 2^+} f(x) = 6 - 2 = 4.$$

Thus, there is no absolute maximum.

Basic Exercise

Draw the graph of a function that attains its maximum value on the interval $[-2, 4]$ but does not have a minimum value. The function does not need to be defined at every point.

Example

Determine the local and global minima and maxima of the function $f(a) = \log_a(e) + \ln(a)$ on the sets

(a) $(0, 1)$

(b) $(2, e)$

(c) $\mathcal{D}(f)$.

Solution

We also convert the first logarithm to natural logarithm to work with the variable a :

$$f(a) = \log_a(e) + \ln(a) = \frac{\ln(e)}{\ln(a)} + \ln(a) = (\ln(a))^{-1} + \ln(a).$$

The derivative:

$$\begin{aligned} f'(a) &= (-1)(\ln(a))^{-2} \cdot \frac{1}{a} + \frac{1}{a} \\ &= \frac{1}{a} \left(1 - (\ln(a))^{-2} \right). \end{aligned}$$

Solution (continued)

The zeros of the derivative:

$$f'(a) = 0$$

$$\frac{1}{a} \left(1 - (\ln(a))^{-2} \right) = 0$$

$$1 - (\ln(a))^{-2} = 0$$

$$(\ln(a))^{-2} = 1$$

$$\frac{1}{(\ln(a))^2} = 1$$

$$\ln(a) = \pm 1$$

$$e^{\ln(a)} = e^{\pm 1}$$

$$a = e^{\pm 1}.$$

The conditions that restrict the derivative f' : $a > 0$ and $\ln(a) \neq 0 \iff a \neq 1$. However, these happen to be the same points where the original function f is not defined:

$$\mathcal{D}(f) = \{a \in \mathbb{R} \mid a > 0 \text{ and } a \neq 1\} = (0, \infty) \setminus \{1\}.$$

Solution (continued)

We will sketch the graph of the function f . A couple of points from the graph:

$$f(e) = \frac{1}{\ln(e)} + \ln(e) = 1 + 1 = 2$$

$$f(e^{-1}) = \frac{1}{\ln(e^{-1})} + \ln(e^{-1}) = \frac{1}{-1 \cdot \ln(e)} + (-1) \ln(e) = -1 - 1 = -2.$$

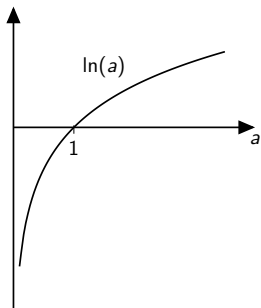


Figure: $\ln(a)$

Solution (continued)

$$\lim_{a \rightarrow 0^+} f(a) = \lim_{a \rightarrow 0^+} \left(\overbrace{\frac{1}{\ln(a)}}^{\rightarrow 0} + \overbrace{\ln(a)}^{\rightarrow -\infty} \right) = -\infty$$

$$\lim_{a \rightarrow 1^-} f(a) = \lim_{a \rightarrow 1^-} \left(\overbrace{\frac{1}{\ln(a)}}^{\rightarrow -\infty} + \overbrace{\ln(a)}^{\rightarrow 0} \right) = -\infty$$

$$\lim_{a \rightarrow 1^+} f(a) = \lim_{a \rightarrow 1^+} \left(\overbrace{\frac{1}{\ln(a)}}^{\rightarrow \infty} + \overbrace{\ln(a)}^{\rightarrow 0} \right) = \infty$$

$$\lim_{a \rightarrow \infty} f(a) = \lim_{a \rightarrow \infty} \left(\overbrace{\frac{1}{\ln(a)}}^{\rightarrow 0} + \overbrace{\ln(a)}^{\rightarrow \infty} \right) = \infty$$

Solution (continued)

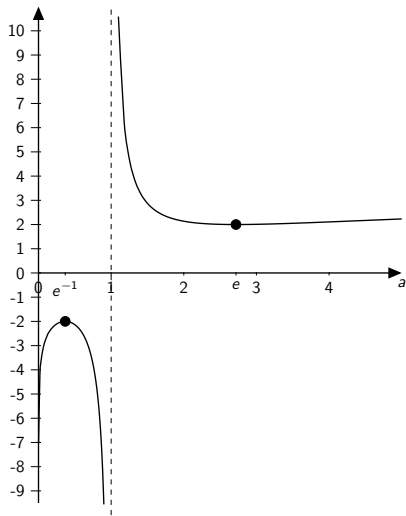


Figure: The extreme values of $f(a) = \log_a(e) + \ln(a)$.

- (a) The global maximum occurs at $a = e^{-1}$, but since the function can take arbitrarily small values in the interval $(0, 1)$, there is no minimum point.
- (b) In the interval $(2, e)$, there is neither a maximum nor a minimum (open interval!).
- (c) The function has a local maximum at $a = e^{-1}$ and a local minimum at $x = e$ throughout its entire domain, but there are no global minima or maxima.

If the endpoints of the interval under consideration are $-\infty$, ∞ , or both, then the corresponding limits $\lim_{x \rightarrow \pm\infty} f(x)$ must be examined. These limits cannot be the extrema of the function (since infinity is not a number), but they may indicate that the function does not have a maximum or minimum value.

Example

Show that a third-degree polynomial has no global extrema.

Solution

It is sufficient to show that an arbitrary third-degree polynomial

$$f(x) = ax^3 + bx^2 + cx + d, \quad a, b, c, d \in \mathbb{R}, \quad a \neq 0,$$

can take arbitrarily large and small values.

Solution (continued)

Based on the graphs, it is evident that they can take such values as the variable grows infinitely large or small, so we calculate the corresponding limits:

$$\begin{aligned}\lim_{x \rightarrow \pm\infty} (ax^3 + bx^2 + cx + d) &= \lim_{x \rightarrow \pm\infty} x^3 \left(a + \underbrace{\frac{b}{x}}_{\rightarrow 0} + \underbrace{\frac{c}{x^2}}_{\rightarrow 0} + \underbrace{\frac{d}{x^3}}_{\rightarrow 0} \right) \\ &= \begin{cases} \pm\infty, & \text{if } a > 0 \\ \mp\infty, & \text{if } a < 0. \end{cases}\end{aligned}$$

Extrema of a Monotonic Outer Function

Basic Exercise

(a) Compare the graphs of the functions $f(x)$ and $\sqrt{f(x)}$ in the following cases:

(i) $f(x) = x^2 + 1$

(ii) $f(x) = \sin(x) + 2$

(iii) $f(x) = e^{x^2}$

(iv) $f(x) = \frac{1}{x^2 + 1}$.

Pay particular attention to the locations of local extrema.

(b) Repeat part (a) for the functions $f(x)$ and $e^{-f(x)}$.

(c) Repeat part (a) when the outer function is some other monotonic (i.e., decreasing or increasing) function than $\sqrt{x}$ or e^{-x} .

(d) Also compare the graphs of the functions $f(x) = \sin(x)$ and $g(x) = \sin^2(x)$.

The previous task illustrates the following result.

Theorem

Let f and g be functions. If f is monotonic, then the composite function $f \circ g$ has the same extreme points as the inner function g (at points where both functions are defined). Furthermore, if f is increasing, then the extrema of the $f \circ g$ and g are of the same type, that is, both minima or both maxima, and if f is decreasing, then the extrema of the $f \circ g$ and g are of the opposite type, that is, the minimum of the inner function corresponds to the maximum of the composite function and the maximum of the inner function corresponds to the minimum of the composite function.

If the outer function is increasing, then the extrema of the composite function and the inner function are of the same type, that is, both minima or both maxima. Conversely, if the outer function is decreasing, then the extrema will be of opposite types, meaning that the minimum of the inner function corresponds to the maximum of the composite function and the maximum of the inner function corresponds to the minimum of the composite function.

If the outer function is not monotonic, then this result does not hold, as demonstrated in part (d) of the previous task.

The previous result can be used to simplify the determination of extrema. This often arises when minimizing some distance, as this typically leads to examining a square root expression (Why?).

Example

Determine the extrema of the function $f(x) = -\ln(x^2 + 1)$.

Solution

The natural logarithm is strictly increasing, so f is strictly decreasing. Thus, the maximum value of function f is found at the minimum point of the inner function $g(x) = x^2 + 1$, and the minimum value at the maximum point of function g . Moreover, since the inner function is positive everywhere, f is defined everywhere, and therefore the extrema are sought in the set $\mathbb{R}$.

Solution (continued)

The graph of function g :

$$g'(x) = 2x,$$

$$g'(x) = 0 \iff 2x = 0 \iff x = 0,$$

$$g'(-1) = -2 < 0,$$

$$g'(1) = 2 > 0.$$

		0		
				x
$g'(x)$		-		+
$g(x)$		$\searrow$		$\nearrow$

Based on the graph, the function g has a global minimum at the point $x = 0$, and there is no global maximum. Thus, the function f has a local maximum at the point $x = 0$, and there is no local minimum. The maximum value is given by

$$f_{\max} = f(0) = \ln(1) = 0.$$

Solution (continued)

From the graph, it can also be concluded that the graph of the function f is of the form:

		0		x
$f'(x)$	+		-	
$f(x)$	$\nearrow$		$\searrow$	

Basic Exercise

Determine the extrema of the functions $f(x) = \sqrt{e^{x^2}}$ and $g(x) = -\sqrt{e^{x^2}}$.

Exercise

Determine which point of the graph of the function $f: \mathbb{R}_{>0} \rightarrow \mathbb{R}$, $f(x) = \frac{1}{x}$, is closest to the origin.